Socially-Weighted Distributed Graph Optimization (SW-DGO) for Autonomous Multi-Agent Service Fleets in Crowded Environments

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***Abstract*—Autonomous service fleets—retail carts, clinical pushchairs, airport trolleys—must route continuously through crowded public space, where classical planners treat pedestrians as obstacles to be cleared rather than people whose personal space carries a cost, and offer no mechanism for symmetric conflicts in single-file corridors. We propose Distributed Dynamic Route Optimization (D^{2}RO), powered by Socially-Weighted Distributed Graph Optimization (SW-DGO): a four-term edge traversal cost $C_i(e,t) = w_D C_{\text{geom}} + w_M C_{\text{mesh}} + w_H C_{\text{social}} + w_S C_{\text{kinematic}}$ minimised *subject to* a directional corridor reservation constraint, solved by incremental D* Lite repair over event-driven V2V telemetry.**

Across $4{,}650$ kinodynamic trials in eleven experiments, each designed so that a claim is attributable to a component, a 2×2 factorial with mesh and reservation disabled in every cell separates social *routing* from reactive *yielding*: the proxemic cost term alone reduces intimate-space exposure by 6.40 person-seconds (95% CI $[-6.48, -6.31]$, $p = 1.5 \times 10^{-9}$) at a cost of 19.03 s, while reactive yielding provides no statistically detectable additional reduction on an already social route ($p = 0.59$) and collapses success to 12% when the proxemic routing cost is absent.

Holding social competence fixed and varying only the coordination architecture then bounds what distribution itself contributes, the question this study is organised around. A human-aware planner carrying the same proxemic cost but no distributed layer matches D^{2}RO's compliance exactly ($p = 1$) while running 8 s faster; mesh and reservation earn their cost only under the topologies our controlled experiments isolate, advancing rerouting by 11.10 s and raising corridor success from 36% to 88%. The unchanged planner transfers across the three simulated topologies evaluated — supermarket, hospital and airport — at 99.0%, 100.0% and 95.0% success. Every table and figure is generated from committed raw data with no hand-entered numbers.

***Index Terms*—Multi-Agent Path Finding (MAPF), Socially-Aware Navigation, Distributed Graph Optimization, Vehicle-to-Vehicle (V2V) Mesh, Human Proxemics, D* Lite, Non-Holonomic Kinematics, Autonomous Mobile Robots (AMRs).**

I. Introduction

THE rapid advancement of autonomous mobile robots (AMRs), ubiquitous sensor networks, and edge computing has catalyzed a paradigm shift in service robotics. While early automated guided vehicles (AGVs) operated primarily within segregated industrial warehouses—isolated from humans behind physical safety barriers and governed by rigid guide-paths—the next generation of autonomous service fleets must operate directly within **human-shared, highly dynamic, and unstructured public environments** [1]–[3].

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Such fleets already appear in several settings. Smart retail shopping trolleys (*Int-Cart*) escort shoppers, assist those with mobility impairments, collect items for online grocery fulfilment, and return themselves from checkout to front-of-store depots [1]. Clinical pushchairs and mobile beds carry patients and supplies between emergency departments, operating suites, radiology, and inpatient wards. Airport luggage trolleys move heavy baggage across open departure concourses, through security screening, and along narrow boarding piers. Library and micro-fulfilment AGVs work in single-file book stacks and storage aisles alongside staff and patrons. What unites them is not the payload but the setting: each shares its floor with people who are under no obligation to get out of the way.

Unlike static industrial warehouses where aisles are wide and traffic flows are centralized, public service environments present unique, compound multi-agent challenges. In these settings, space is severely constrained by orthogonal shelving fixtures, corridors are frequently single-file, and human pedestrians move stochastically with complex browsing and loitering behaviors [2].

A. Core Research Motivations & Technical Bottlenecks

When deployed in complex, fixture-dense public service environments, traditional Multi-Agent Path Finding (MAPF) and reactive obstacle avoidance algorithms suffer from four fundamental technical bottlenecks:

1) The Geometry-Kinematic Disconnect in Concave Fixtures (The ORCA Trap): Classical continuous collision avoidance methods—such as Optimal Reciprocal Collision Avoidance (ORCA) [4] and Artificial Potential Fields (APF)—rely on local velocity-space half-planes or repulsive force vectors. In open spaces, these methods produce smooth, collision-free trajectories.

However, in realistic retail and hospital layouts filled with orthogonal $90°$ shelf corners, end-cap displays, and narrow U-bays, reactive methods are prone to **local potential minima**. When a robot encounters a pedestrian near a shelf corner, the repulsive force from the human and the repulsive vector from the orthogonal wall can cancel the goal attraction vector ($\|\mathbf{F}_{\text{net}}\| \to 0$), stalling the agent until the pedestrian moves on.

We stress that this is a *delay* mechanism, not necessarily a failure mechanism. Our measurements (Section VI-B) show that, given a time budget calibrated to the true mission duration, an artificial potential field completes every mission it is given; what it cannot do is complete them without repeatedly driving through people. The deficiency of reactive avoidance in these settings is therefore social and temporal rather than a simple inability to reach the goal, and the contribution of this paper should be read against that corrected baseline.

2) Social Blindness of Static Shortest-Path Solvers (The A^* Flaw): Traditional discrete graph planners (e.g., Static A^*, Dijkstra) compute global paths based strictly on static Euclidean distance $D(u, v)$ [5]. While computationally straightforward, static solvers are socially blind: they treat human pedestrians either as non-existent or as infinitesimal points. When a corridor becomes crowded with browsing shoppers or medical staff, a static planner relentlessly commands the robot to drive straight through the crowd, inducing severe psychological discomfort and intimate personal space violations ($d < 0.8$ m).

3) Symmetrical Live-Locks in Single-File Corridors: In narrow architectural passages (e.g., supermarket grocery aisles, clinical hospital corridors) whose width does not admit two carts side by side with the required clearances (Eq. (12)), bidirectional passing is not possible. When two opposing agents enter a single-file aisle simultaneously, reactive avoidance algorithms oscillate in place, producing permanent symmetrical live-locks [6].

4) Information Isolation and Delayed Fleet Backtracking: In decentralized multi-agent systems without communication, robots operate with myopic, onboard sensing radius. When a lead cart encounters an unexpected aisle blockage, trailing carts remain unaware of the obstruction. Trailing units travel all the way to the blocked entrance before sensing the congestion locally, forcing complete reversals and causing severe inflation in fleet makespan [7].

B. Target Application Fields

The framework targets the same four domains, each of which stresses it differently. Retail supermarkets combine narrow grocery aisles with arterial promenades (**Action Alley**), end-caps and checkout queues. Clinical facilities impose a priority hierarchy: emergency trauma vehicles hold right-of-way while routine pushchairs yield into **Turnout Alcoves** (V_{alcove}). Aviation terminals pair wide open concourses with security pinch points and elongated gate piers. Public facilities and micro-fulfilment stacks offer the tightest physical clearances of all. Three of these—supermarket, hospital and airport—are implemented as simulation domains and evaluated in Section VI-F.

C. The Proposed Solution: The D^{2}RO Framework

To overcome these bottlenecks, this paper proposes the **Distributed Dynamic Route Optimization (D^{2}RO)** framework powered by **Socially-Weighted Distributed Graph Optimization (SW-DGO)**. Rather than treating navigation as isolated reactive avoidance or purely local search, D^{2}RO establishes a **distributed, time-varying graph-cost field** $C_i(e, t)$ shared across peer agents:

$$C_i(e,t) = w_D \cdot C_{\text{geom}}(e) + w_M \cdot C_{\text{mesh}}(e,t) + w_H \cdot C_{\text{social}}(e,t) + w_S \cdot C_{\text{kinematic}}(i,e,t) \quad \text{subject to } R_{\text{lock}} \tag{1}$$

seamlessly unifying incremental heuristic graph repair (D* Lite) [8] with distributed anticipatory cost propagation, continuous Gaussian proxemics [2], [3], directional corridor reservation protocols (R_{lock}), and non-holonomic vehicle clearance envelopes.

D. Research Aims & Key Contributions

The framework is designed against four explicit aims, and we return to each of them in the discussion and conclusion. **A1, social compliance:** route service fleets so that pedestrians' intimate space is not entered, accepting bounded additional travel time as the price (Section VI-C). **A2, distributed coordination without a central server, and the conditions under which it is warranted:** resolve congestion and single-file conflicts using only peer-to-peer information within a bounded sensing radius, and establish when that layer earns its cost by comparing it against a non-distributed planner of equal social competence rather than against a socially blind one (Section VI-H). **A3, architectural generality and real-time feasibility:** operate unchanged across dissimilar public topologies within a 50 ms control budget (Sections VI-F and VI-G). **A4, attributable evaluation:** measure the contribution of the routing policy *itself*, separately from the vehicle controller that executes it, and establish whether the result depends on the particular weights chosen (Sections VI-C and VI-I).

The primary scientific contributions of this research are:

1) **Distributed Anticipatory Edge-Cost Propagation & Horizon Extension:** We formalize a distributed dynamic graph-cost field in which localized physical and social perturbations observed by leading agents become peer-to-peer, time-decayed edge penalties $C_{\text{mesh}}(e, t)$. This extends each robot's effective planning horizon ($\mathcal{O}_i^{\text{effective}}(t)$) beyond its onboard sensing radius (R_s). Under a controlled experiment that isolates the mechanism (Section VI-H), agents divert 10.70 ± 4.20 s before they could have observed the obstruction directly, and more than halve their backtracking distance (1.08 ± 0.68 m vs. 2.73 ± 0.87 m, Holm-adjusted $p = 3.2 \times 10^{-8}$).
2) **Distributed Directional Bottleneck Reservation Protocol:** We formalize an explicit distributed protocol for single-file topological bottlenecks (Eq. (12)) via directional reservation tuples $\mathcal{L}_e = \langle \text{owner}, \vec{d}, t_{\text{acquire}}, t_{\text{expire}}, \text{priority} \rangle$ and Turnout Alcoves (V_{alcove}). In the controlled corridor experiment the protocol raises mission success from 36.0% to 88.0% ($p = 1.7 \times 10^{-4}$). The mechanism operates by *cost-projected diversion* rather than by mutual exclusion: agents leave the contested corridor (2.16 ± 1.36 off-corridor vertices against 0.00, $p = 2.3 \times 10^{-8}$) while accumulating 0.030 ± 0.070 s of waiting. We therefore describe it as a directional reservation rather than a mutex, and analyse the distinction in Section VI-H.
3) **Kinodynamically & Socially Conditioned Incremental Optimization:** We unify asymmetric human proxemic discomfort fields (C_{social}) and vehicle-specific clearance envelopes ($C_{\text{kinematic}}$) directly into an incremental D* Lite graph repair engine, executing incremental repairs

at a median of 0.002 ms (p95 0.10 ms) without global re-heapification. Ablation shows the proxemic term is load-bearing: removing it collapses mission success from 100.0% to 11.0%.

4) **A Boundary Condition on Distributed Coordination:** We hold social competence fixed and vary only the coordination architecture, which is what makes the distributed layer's contribution separable. Measured this way it is not a general improvement. In a broad randomized scenario it contributes nothing detectable: an ordinary human-aware D^* Lite planner carrying the same proxemic cost but no mesh and no reservation matches the full system's social compliance exactly ($p = 1$) while completing missions 8.12 s sooner. Under the two topologies the mechanisms are designed for — a blockage outside a follower's sensing range, and a single-file corridor entered from both ends — the same layer is decisive, advancing rerouting by 10.70 s and raising corridor success from 36.0% to 88.0%. We report this boundary as a finding rather than as a caveat, because it is the practically useful part: it tells a deployer which topologies justify the radio and which do not.

5) **Comprehensive Multi-Domain Empirical Benchmark Suite:** We validate the framework across three distinct architectural topologies (Retail Supermarket, Clinical Hospital, and Airport Terminal) 4,650 simulation trials in eleven experiments, including a matched-controller baseline that separates routing from vehicle dynamics, a local-social baseline that separates the distributed layer from ordinary human-aware navigation, a route-by-yield factorial, a four-weight sensitivity study on a disjoint seed set, and communication degradation applied where the mesh is causally responsible. Every table and figure in this paper is generated programmatically from the committed raw data by a single analysis pipeline, with no hand-entered numbers; datasets and code are released under open licenses.

II. Related Work & Literature Review

The proposed D^2RO framework builds upon foundational work across Multi-Agent Path Finding (MAPF), dynamic graph replanning, local collision avoidance, ad-hoc mesh communication, physical trolley mechatronics, and human-aware navigation.

A. Decentralized and Lifelong MAPF

Stern *et al.* [5] define the classical MAPF problem and its variants, establishing the fundamental vertex and edge conflict models used in grid environments. To address continuous operations, Ma *et al.* [9] introduced Lifelong Multi-Agent Path Finding for Online Pickup and Delivery (MAPD), where agents dynamically receive tasks without resting.

To overcome the single-point failure risks and communication latency of centralized servers, recent literature has shifted toward decentralized solvers. Dergachev and Yakovlev [10] explored decentralized unlabeled MAPF using target and priority swapping, while Keskin *et al.* [11] presented a decentralized MAPF framework utilizing automated negotiation protocols. Furthermore, learning-based approaches have gained traction: Sartoretti *et al.* [12] introduced PRIMAL, utilizing reinforcement and imitation learning for decentralized pathfinding, and Skrynnik *et al.* [13] developed "Learn to Follow" to separate global heuristic sub-goal allocation from low-level local policies. Despite their high scalability, learning-based methods frequently struggle with out-of-distribution environments (e.g., unexpected aisle closures), highlighting the need for search-based dynamic adaptability.

B. Dynamic Replanning in Unknown and Stochastic Environments

In retail environments, the topological graph changes dynamically as aisles become blocked or crowded. Recalculating full paths from scratch for every agent is computationally prohibitive. Koenig and Likhachev [8] addressed this with D^* Lite, an incremental heuristic search algorithm that recalculates only the segments of a path affected by dynamic edge-cost changes.

The application of D^* Lite to multi-agent systems was successfully demonstrated by Al-Mutib *et al.* [14], who utilized it for real-time path planning by treating peer paths as temporary obstacles. Additionally, Wagner and Choset [15] developed M^*, dynamically varying the dimensionality of the search space only when agent paths conflict. D^2RO adapts these incremental principles, allowing trolleys to dynamically inflate traversal costs based on real-time congestion data without recomputing the entire global map.

C. Kinematic Coordination and Local Collision Avoidance

While MAPF and D^* Lite provide global waypoints, continuous kinematic control is required for safe micro-maneuvers when agents cross paths. Van den Berg *et al.* [4] introduced Optimal Reciprocal Collision Avoidance (ORCA), providing sufficient conditions for multiple robots to avoid collisions in continuous space without explicit communication.

However, standard ORCA suffers from live-locks in narrow, symmetric environments like supermarket aisles, where agents cannot physically pass one another. Dergachev and Yakovlev [6] specifically address this, proposing a system that uses continuous reciprocal collision avoidance but falls back on a locally confined MAPF instance when a deadlock is detected. D^2RO leverages this exact synthesis: relying on continuous reactive models for open spaces, and spatiotemporal edge reservations for single-file corridors.

D. Multi-Robot Ad-Hoc Communication Protocols

The transition from centralized control to a truly distributed D^2RO system requires robust peer-to-peer communication. Gielis *et al.* [7] emphasize the critical need for co-designing robotic planning algorithms alongside network constraints.

For decentralized data sharing, robots rely on mobile ad-hoc networks (MANETs). Slyusar and Kulich [16] evaluated routing protocols for MANETs in multi-robot exploration. Additionally, Loems [17] investigated robot communication within Swarm

SLAM, demonstrating how independent agents merge local spatial data over a distributed mesh. In D^2RO, this translates to an event-driven telemetry protocol where agents broadcast localized edge-cost penalties across a V2V mesh, allowing distant agents to proactively reroute.

E. Indoor Positioning and Physical Hardware Implementation

Unlike simulated grids, physical shopping trolleys require absolute spatial grounding and customized mechatronics. Zafari *et al.* [18] provide a comprehensive survey of indoor localization technologies, highlighting the superiority of Ultra-Wideband (UWB) and BLE for centimeter-level accuracy in GPS-denied environments. Clark *et al.* [19] expanded on this with the TEAM framework, demonstrating effective trilateration and mapping utilizing a localized robotic network, while Nugraha *et al.* [20] proved that fusing Indoor Positioning Systems (IPS) with wheel odometry via Extended Kalman Filters (EKF) drastically reduces navigation drift.

Bringing these concepts into physical retail spaces, Mohamad Azlan *et al.* [1] developed the *Int-Cart*, an autonomous mobile trolley robot. Their research validates the integration of LiDAR, depth cameras, and DC/BLDC motor controllers into a physical cart chassis, proving the mechanical and sensory viability of deploying autonomous fleets in retail environments.

F. Human-Aware Navigation & Research Gap

A consistent theme of the human-aware navigation literature is that robots cannot treat people simply as "moving cylindrical obstacles". Kruse *et al.* [2] survey this position at length: comfort, legibility and social convention are objectives in their own right, not by-products of collision avoidance, and a planner that satisfies only the geometric constraint can still be experienced as intrusive. Planners must therefore incorporate proxemics (personal-space boundaries) and contextual human activity into routing itself.

The Research Gap: The two bodies of work reviewed above are largely disjoint, and the gap lies between them. The decentralized MAPF and ad-hoc coordination literature (Sections II-A–II-D) optimises throughput, completion and conflict resolution among robots, and its baselines are other planners — centralized solvers, reactive avoiders, learned policies — none of which prices human personal space. The human-aware navigation literature (this section) prices personal space carefully but is predominantly single-robot, so it has no distributed layer whose contribution could be isolated.

Consequently a distributed coordination layer is customarily evaluated against a baseline that lacks *both* the distribution *and* any social competence, and that comparison cannot separate the two: a socially weighted distributed planner will outperform a socially blind one whether or not the distribution does any work. The necessity of the distributed layer is thereby assumed rather than shown.

The gap we address is therefore not the absence of another hybrid. It is the absence of a study that holds social competence *fixed* and varies only the coordination architecture, and so can say when a distributed layer is warranted at all. This paper is organised around that question. We build the hybrid framework, and then spend most of the evaluation determining whether its distributed half is necessary — including against a baseline constructed to be its equal in social behaviour while sharing none of its coordination machinery. The answer is conditional, and the conditions are the contribution.

III. Problem Formulation & The SW-DGO Framework

A. Topological Roadmap and Kinematic Model

Let the indoor facility floorplan be represented by an **undirected physical roadmap** graph $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_n\}$ denotes navigable topological waypoints (e.g., aisle intersections, passing bays, docking bays) and $E \subseteq V \times V$ represents physical corridor edges. From this undirected physical structure, each agent i maintains a **directed local cost graph** $\vec{G}_i = (V, A_i)$, whose directed arcs $a = (u, v) \in A_i$ carry directional traversal penalties $C_i(u, v, t)$ and directional corridor lock states.

Each autonomous service cart $i \in \{1, \ldots, N_{\text{carts}}\}$ is governed by non-holonomic unicycle kinematics:

$$\begin{aligned} \dot{x}_i(t) &= v_i(t) \cos\theta_i(t), \\ \dot{y}_i(t) &= v_i(t) \sin\theta_i(t), \quad \dot{\theta}_i(t) = \omega_i(t) \end{aligned} \tag{2}$$

subject to physical actuator bounds $|v_i(t)| \leq v_{\max}$ and $|\omega_i(t)| \leq \omega_{\max}$.

B. The Distributed Dynamic Edge-Cost Field Formulation

Rather than relying on centralized graph allocation or isolated myopic sensing, SW-DGO establishes a **distributed, time-varying edge-cost field** $C_i(u, v, t)$ maintained locally by each autonomous agent i. The edge traversal cost $e = (u, v) \in E$ is formalized as:

$$\begin{aligned} C_i(u, v, t) = w_D \cdot C_{\text{geom}}(u, v) + w_M \cdot C_{\text{mesh}}(u, v, t) \\ + w_H \cdot C_{\text{social}}(v, t) + w_S \cdot C_{\text{kinematic}}(i, v, t) \end{aligned} \tag{3}$$

minimised over paths *subject to* the directional reservation constraint

$$e \notin \mathcal{E}_i^{\text{reserved}}(t) \tag{4}$$

where $\mathcal{E}_i^{\text{reserved}}(t)$ is the set of single-file edges currently claimed by a peer whose claim precedes agent i's in the total order of Section III-B4.

We state reservation as a **constraint rather than a weighted term**, and this is a correction to earlier presentations of this framework, which wrote it as $w_R \cdot C_{\text{mutex}}$ and described w_R as a fifth tunable weight. That formulation was not faithful to the mechanism. C_{mutex} takes only the values 0 and ∞: a reserved edge is unavailable, and no finite coefficient makes it more or less unavailable, while an unreserved edge contributes $w_R \cdot 0 = 0$ for every w_R. The coefficient therefore had no operative magnitude in any reachable state, which is precisely why perturbing it moves no measured quantity (Section VI-I). Feasibility is the honest description, and the framework is accordingly **four weighted soft terms plus one hard constraint**.

The remaining dimensionless weights $\mathbf{w} = [w_D, w_M, w_H, w_S] = [1.0, 1.5, 2.0, 1.2]$ are *hand-selected nominal values*, not the output of a calibration procedure; Section VI-I reports how the system behaves when each is perturbed. They balance physical progress against social proxemics, collaborative telemetry and vehicle clearance:

- $w_D = 1.0$: Baseline Euclidean physical distance (C_{geom} in meters).
- $w_M = 1.5$: Weights collaborative V2V congestion alerts $1.5\times$ higher than metric distance, ensuring trailing carts proactively reroute through clear parallel corridors rather than deadheading into blocked aisles.
- $w_H = 2.0$: Prioritizes pedestrian personal-space comfort $2.0\times$ above travel distance, enforcing compliant yielding and wide social berths.
- $w_S = 1.2$: Kinetic clearance penalty preventing chassis corner scrapes ($1.2\times$ baseline) and anti-tailgating violations.

1) Intrinsic Metric Geometry $C_{geom}(u,v)$:

$$C_{\text{geom}}(u,v) = \|\mathbf{p}_u - \mathbf{p}_v\|_2 \tag{5}$$

A turn penalty $\alpha_{\text{turn}}|\Delta\theta|$ would be a natural addition here, and earlier presentations of this formulation included one. We have removed it, because charging for a heading change requires the search state to *carry* a heading: the cost of entering an edge then depends on the edge traversed before it, which a position-indexed graph cannot express. Implementing it correctly means augmenting the state space with orientation and re-deriving the incremental repair over that larger space. We record it as future work (Section VII-D) rather than display a term the evaluated system does not contain.

Cornering is not thereby ignored: the motion layer applies a bounded angular rate $\omega_{\max} = 2.50$ rad/s and decelerates through turns, so sharp direction changes are penalised in *execution* even though they are not priced in the *search*. The distinction matters for interpreting the results: D^{2}RO does not prefer straighter routes, it merely traverses crooked ones more slowly.

2) Distributed Anticipatory Cost Propagation & Horizon Extension $C_{mesh}(u,v,t)$: In decentralized multi-agent systems, myopic range-limited sensing restricts an agent's local obstacle horizon to $\mathcal{O}_i^{\text{local}}(t) = \{\mathbf{x} \mid \|\mathbf{x} - \mathbf{p}_i(t)\| \leq R_s\}$. To eliminate the severe makespan penalty of deadheading into remote blockages, SW-DGO formalizes an **effective perception horizon extension**:

$$\mathcal{O}_i^{\text{effective}}(t) = \mathcal{O}_i^{\text{local}}(t) \cup \bigcup_{j \in \mathcal{N}_i} \mathcal{O}_j(t - \tau_{ij}) \tag{6}$$

When agent j discovers a congestion event at edge (u,v) at timestamp t_k, it broadcasts a localized event-driven telemetry packet over the ad-hoc mesh with bounded time-to-live (TTL = 3). The received penalty decays exponentially over time:

$$C_{\text{mesh}}(u,v,t) = \sum_{k \in \mathcal{M}(u,v,t)} \gamma_k \cdot \exp\left(-\lambda_{\text{decay}}(t - t_k)\right) \tag{7}$$

where γ_k is the initial congestion penalty amplitude and λ_{decay} is the temporal decay rate, guaranteeing that stale congestion information naturally attenuates without permanently poisoning the routing topology.

3) Continuous 2D Asymmetric Human Proxemics $C_{social}(v,t)$: To model human psychological personal space (Hall's Proxemics), we formulate an **asymmetric directional Gaussian potential field** aligned with pedestrian heading θ_h:

$$H_{\text{prox}}(\mathbf{p}, \mathbf{h}_j, \theta_h) = A_j \cdot \exp\left(-\frac{1}{2}\left[\left(\frac{x_{\text{local}}}{\sigma_x}\right)^2 + \left(\frac{y_{\text{local}}}{\sigma_y}\right)^2\right]\right) \tag{8}$$

where relative coordinates are transformed into the pedestrian's local body frame:

$$\begin{bmatrix} x_{\text{local}} \\ y_{\text{local}} \end{bmatrix} = \begin{bmatrix} \cos\theta_h & \sin\theta_h \\ -\sin\theta_h & \cos\theta_h \end{bmatrix} \begin{bmatrix} x - x_j \\ y - y_j \end{bmatrix} \tag{9}$$

with asymmetric longitudinal variance reflecting expanded front personal space:

$$\sigma_x = \begin{cases} 1.35 \text{ m}, & x_{\text{local}} \geq 0 \text{ (front)} \\ 0.60 \text{ m}, & x_{\text{local}} < 0 \text{ (rear)} \end{cases} \qquad \sigma_y = 0.90 \text{ m} \tag{10}$$

The continuous social edge penalty is integrated along the edge trajectory segment:

$$\begin{aligned} C_{\text{social}}(u,v,t) &= \int_e H_{\text{prox}}(\mathbf{p}, t)\, ds \\ &= \|\mathbf{p}_v - \mathbf{p}_u\| \int_0^1 H_{\text{prox}}\big((1-\tau)\mathbf{p}_u + \tau\mathbf{p}_v, t\big)\, d\tau \end{aligned} \tag{11}$$

4) The Directional Bottleneck Reservation Constraint *$C_{mutex}(u,v,t)$:*

a) When is a passage single-file?: Two rectangular carts of width $W_{\text{cart}} = 0.48$ m can pass one another in a straight corridor only if the corridor admits both bodies side by side together with a wall clearance c_w on each side and a mutual clearance c_m between them:

$$W_{\text{corridor}} \;\geq\; 2W_{\text{cart}} + 2c_w + c_m. \tag{12}$$

With the clearance margin used throughout this work ($c_w = c_m = 0.54$ m, Table I) the threshold is 2.58 m. For two *discs* of radius r the corresponding condition is $W \geq 4r$, not $2r$: each centre must stay r from its wall and the centres must be $2r$ apart. Earlier presentations of this work stated the condition as $W_{\text{corridor}} < 2r_{\text{safety}}$, which is geometrically incorrect in either formulation, and we correct it here.

We are also explicit that 0.40 m is an **effective safety radius**, chosen to bound the swept envelope of a turning cart, and not the inscribed radius of the 0.72×0.48 m chassis, whose inscribed radius is 0.24 m. The earlier label was simply wrong.

b) Which edges are designated single-file.: Applying Eq. (12), the 2.1 m grocery aisles fall below the 2.58 m threshold and are designated single-file, as are the clinical corridors; the promenades and concourses are not. We note plainly that a 2.1 m aisle would admit two 0.48 m carts on bare geometry alone — the designation follows from the operational clearance the fleet is required to maintain, and from the fact that retail aisles are routinely occupied by standing shoppers, end-cap overhang and parked trolleys, not from chassis width in isolation. The designation is a property of the environment definition and is fixed per edge; it is not recomputed at run time.

TABLE I: System Simulation Parameters & Physical SI Metric Calibration.

Parameter Description	Symbol	Physical Value (SI)
Spatial Resolution Scale	s_{res}	1 px = 0.03 m (1 m = 33.33 px)
Physics Integration Rate	Δt	0.05 s (20 Hz control loop)
GUI Rendering Rate	f_{gui}	60 FPS (16.6 ms interpolation)
Chassis Dimensions	$L \times W$	0.72 m × 0.48 m (24 × 16 px)
Effective safety / swept-envelope radius	r_{robot}	0.40 m (13.3 px)
Maximum Linear Speed	$v_{\max}$	1.20 m/s (40.0 px/s)
Maximum Angular Rate	$\omega_{\max}$	2.50 rad/s
Shelf Clearance Margin	d_{shelf}	0.54 m (18.0 px)
Following Gap Distance	d_{follow}	1.08 m (36.0 px)
V2V Comm Range	R_{mesh}	10.50 m (350.0 px)
V2V Packet Size	S_{pkt}	64 Bytes
Mesh Hop Limit	TTL	3 hops
Proxemic Front Space	σ_{front}	1.35 m (45.0 px)
Proxemic Lateral Space	σ_{side}	0.90 m (30.0 px)
Proxemic Rear Space	σ_{rear}	0.60 m (20.0 px)
Onboard Sensing Radius	R_s	7.20 m (240.0 px)
Arrival Tolerance (all planners)	r_{arrive}	0.84 m (28.0 px)

c) The reservation itself.: Conflicts in such passages are resolved by a directional reservation tuple:

$$\mathcal{L}_e(t) = \langle \text{AgentID}, \vec{d}_{\text{traversal}}, t_{\text{acquire}}, t_{\text{expire}}, \text{Priority} \rangle \tag{13}$$

When agent i claims edge (u, v), the reverse edge cost for opposing agents is set to:

$$C_{\text{mutex}}(v, u, t) = \begin{cases} \infty, & (u, v) \text{ reserved by peer } j \neq i \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

An opposing agent observes this infinite cost and, because the reservation enters its *graph* rather than its controller, D* Lite reroutes it through a parallel aisle or a Turnout Alcove (V_{alcove}) instead of halting at the corridor mouth. Section VI-H shows empirically that this diversion—rather than queueing—is the operative behaviour.

5) Kinetic Chassis Safety Clearance Envelope $C_{kinematic}(i, v, t)$: This term has two components, and we write both because the implementation applies both. A *static* contribution prices proximity to fixed geometry, and a *dynamic* contribution prices proximity to peer vehicles:

$$C_{\text{kinematic}}(i, \mathbf{p}, t) = \underbrace{\kappa\big(d_{\text{shelf}} - d(\mathbf{p}, \mathcal{S})\big)_+}_{\text{static fixture clearance}} + \underbrace{\sum_{j \neq i} A_{\text{trolley}} \exp\left(-\frac{\|\mathbf{p} - \mathbf{p}_j(t)\|^2}{2\sigma^2_{\text{trolley}}}\right)}_{\text{dynamic peer envelope}} \tag{15}$$

where $d(\mathbf{p}, \mathcal{S})$ is the distance from $\mathbf{p}$ to the nearest shelf face, $(\cdot)_+$ denotes the positive part, and κ scales the clearance deficit. The static component enforces the 0.54 m shelf clearance buffer; the dynamic component enforces the 1.08 m anti-tailgating gap. Earlier presentations displayed only the peer term, which understated the component whose ablation produces the largest change in fixture contact (Section VI-C).

C. Comparison with Alternative Simulation Paradigms

To contextualize the architectural design of D²RO, we contrast it with four standard robotics simulation paradigms:

Algorithm 1 Socially-Weighted Distributed Graph Optimization (SW-DGO)

Require: Roadmap graph $G = (V, E)$, Start s_{start}, Goal s_{goal}, V2V Mesh $\mathcal{N}$
Initialize local D* Lite planner with G and s_{goal}
Compute initial shortest path $\Pi \leftarrow$ ComputeShortestPath()
while $s_{\text{current}} \neq s_{\text{goal}}$ **do**
  $\Delta\mathcal{M} \leftarrow$ FetchInboundMeshPackets($\mathcal{N}$)
  cost_changed $\leftarrow$ **false**
  for each packet $pkt \in \Delta\mathcal{M}$ **do**
    $(u, v) \leftarrow pkt$.edge
    Update edge cost: $c(u, v) \leftarrow C(u, v, t)$ via Eq. (3)
    D*Lite.NotifyEdgeCostChange(u, v)
    cost_changed $\leftarrow$ **true**
  end for
  Sample human positions $\{h_j\}$ within radius $R_{\text{sense}} =$ 240 px
  if ProxemicsChanged($\{h_j\}$) **then**
    Update $H_{\text{prox}}(u, v, t)$ for local edges
    cost_changed $\leftarrow$ **true**
  end if
  if cost_changed **then**
    D*Lite.ComputeShortestPath()
    $\Pi \leftarrow$ D*Lite.ExtractPath()
  end if
  $s_{\text{next}} \leftarrow$ GetNextWaypoint(Π)
  if EdgeIsSingleFile($s_{\text{current}}, s_{\text{next}}$) **then**
    AcquireCorridorLock($s_{\text{current}}, s_{\text{next}}$)
    Broadcast(LOCK_REQUEST($s_{\text{current}}, s_{\text{next}}$))
  end if
  Execute non-holonomic step toward s_{next} with S_{trolley} envelope
  Decay active mesh penalties: $W_{\text{mesh}} \leftarrow W_{\text{mesh}} \cdot e^{-\lambda \Delta t}$
end while
return Mission Complete

1) **Classical 2D Grid-World MAPF (CBS, PBS, EECBS):** Assumes synchronous discrete time steps and point holonomic agents. Ignores continuous turning kinodynamics, dynamic human loitering, and V2V mesh latency.
2) **Pure Continuous Multi-Agent Navigators (ORCA, Social Force):** Operates in continuous velocity space without topological roadmaps. Lacks global topological awareness, so an agent driven into an infeasible velocity state in a narrow corridor has no route-level recourse. Our implementation exhibits this mode (Section VI-B), though we do not generalise from it to the published algorithm.
3) **Network-Centric Packet Simulators (NS-3, OMNeT++):** Models high-fidelity RF propagation and MAC collisions, but lacks robot kinodynamic physics and graph search.
4) **3D Rigid-Body Physics Engines (Gazebo, Isaac Sim):** Models full multi-body dynamics, contact mechanics, and sensor ray-tracing. However, executing $N = 100$ randomized Monte Carlo trials across multi-agent fleets

is computationally prohibitive ($> 100\times$ slower than real time).

5) **The SW-DGO Hybrid Approach (The Goldilocks Paradigm):** Couples non-holonomic unicycle kinodynamics ($dt = 0.05$ s, 60 FPS) with continuous Gaussian proxemics, an ad-hoc V2V mesh simulator, and incremental D* Lite graph repair, providing deterministic, real-time evaluation (< 0.15 ms).

IV. Multi-Domain Simulation Topologies

The framework is validated across three architecturally divergent environments, chosen because they stress a social planner in different ways rather than because they differ cosmetically.

The **retail supermarket** (36×24 m) is fixture-dense: six parallel grocery aisles of width $W_{\text{aisle}} = 2.1$ m, too narrow for two carts to pass, connected by three transverse promenades (Back Promenade, Action Alley, Front Concourse) with cashier registers and front-of-store return depots. Routing here is a choice among a small number of parallel corridors.

The **clinical hospital** shares those dimensions but adds a priority structure: Emergency Trauma (ER), sterile operating and MRI suites, inpatient wards, and designated **Turnout Alcoves** (V_{alcove}) set into single-file corridors so that a routine pushchair can pull aside and grant clearance to an emergency vehicle. Its corridors are the narrowest of the three, but the topology is strongly channelised, so there is little to decide once a route is committed.

The **airport terminal** inverts that: a large open departure plaza with check-in banks, narrow security screening chokepoints, and elongated boarding gate piers (Gates A1–A4, B1–B4). Open space means many admissible routes, and Section VI-F shows this abundance of choice — not scarcity — is what makes it the hardest domain for the planner.

V. Spatial Proxemic Heatmaps & Spatiotemporal Analysis

A. Social Detour Heatmap Analysis

Fig. 1 maps the continuous 2D Gaussian discomfort field $H_{\text{prox}}(x, y)$ alongside executed trajectories. Static A^* (dashed red) charges directly through the dense crowd in Aisle 3, whereas D^{2}RO (solid blue) executes a proactive detour along Action Alley and Aisle 2, achieving zero intimate personal space violations.

B. Spatiotemporal Turnout Alcove Resolution

Fig. 2 illustrates the longitudinal space-time trajectory (X, t) in clinical corridors. Emergency Pushchair P_1 maintains full velocity under priority lock ($R_{\text{lock}} = \infty$), while routine Pushchair P_2 pulls into the Turnout Alcove at $X = 650$ px ($t = 4.5$ s $\rightarrow$ 10.5 s), resolving the conflict without either agent reversing. The figure illustrates intended behaviour; the quantitative evidence is in Section VI-H.

C. Airport Concourse Flow Streamlines

Fig. 3 demonstrates 2D velocity streamlines around high-density passenger clusters in open airport concourses, highlighting smooth non-holonomic curvature without erratic stopping.

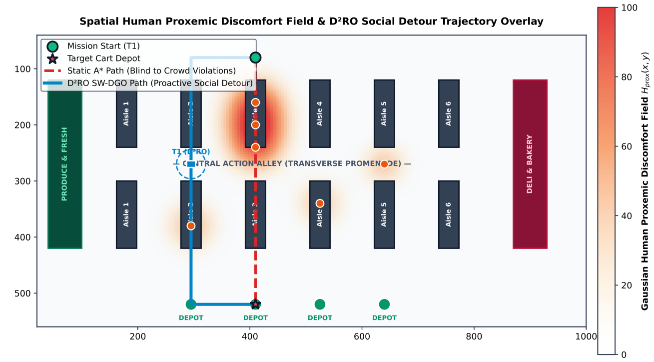

Fig. 1: Continuous 2D Gaussian Discomfort Field $H_{\text{prox}}(x, y)$ heatmap and trajectory overlay comparing Static A^* (blind path through Aisle 3 crowd) against D^{2}RO proactive social detour along Action Alley.

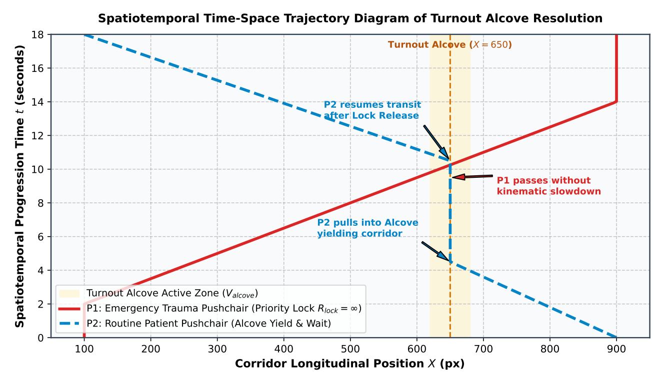

Fig. 2: Spatiotemporal time-space trajectory diagram of Turnout Alcove resolution: Emergency Pushchair P_1 maintains full velocity under priority lock $R_{\text{lock}} = \infty$, while routine Pushchair P_2 yields inside the alcove bay.

VI. Experimental Results & Discussion

A. Experimental Setup & Randomization Protocol

All experiments were conducted using the decoupled 2D kinodynamic simulation framework integrating non-holonomic unicycle dynamics ($\Delta t = 0.05$ s, 20 Hz control loop) across 1200×800 px (36.0 m $\times$ 24.0 m) floorplans. To ensure statistical rigor and reproducibility:

- **Random Seed Policy:** Each evaluation executed $N = 100$ independent Monte Carlo trials seeded deterministically. Crucially, every planner in a given trial receives the *same* seed, so comparisons are **seed-paired** rather than independent: the pedestrian crowd a baseline faces is the crowd D^{2}RO faces.
- **Dynamic Pedestrian Behavior:** Pedestrians move with stochastic velocities $v_h \in [0.8, 1.2]$ m/s, random waypoint heading deflections, and a 30% browsing loitering probability ($p_{\text{loiter}} = 0.30$) generating dynamic Gaussian discomfort Halos.
- **Common Completion Criterion:** Within any given experiment all planners share an identical arrival tolerance

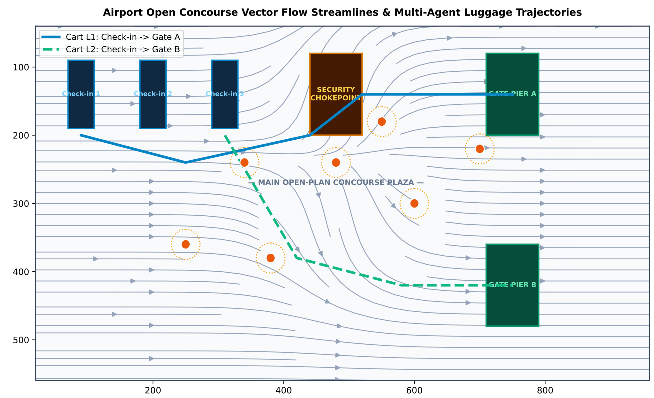

Fig. 3: Airport open concourse vector flow streamlines and multi-agent luggage cart trajectories smoothly navigating around high-density passenger clusters.

of 0.84 m (approximately one cart length) and an identical time budget. An earlier revision applied planner-specific tolerances, scoring the proposed method against a $2.3\times$ more lenient criterion than its baselines; that asymmetry is removed and every result reported here uses the common criterion. Budgets are set well above the longest observed mission for the experiment class so that a timeout indicates genuine non-completion rather than a truncated run: $T_{\max} = 180$ s for the benchmark, ablation, cross-domain and crowd-density experiments, 240 s for fleet-size scaling (where queueing grows with fleet size), and 120 s for the two-cart mechanism experiments.

- **Statistical Analysis:** Because the trials are seed-paired and several outcome distributions are zero-inflated and strongly right-skewed, we do not assume normality. Continuous outcomes are compared with the **Wilcoxon signed-rank** test on within-pair differences, supported by paired bootstrap confidence intervals; binary mission success is compared with **McNemar's exact** test. p-values are **Holm-adjusted** within each family of comparisons, and every reported p belongs to one specific outcome—no single significance claim is attached to several unrelated metrics. Skewed outcomes are summarised as median [IQR]; symmetric ones as $\mu \pm \sigma$.

B. Physical Insights and Failure Modes Analysis

1) Artificial Potential Fields: a social failure, not a navigational one: An earlier revision of this study reported that APF achieves 0.0% mission success and built a local-minima argument upon it. Under the common completion criterion and a time budget calibrated to true mission durations, that result does not reproduce: **APF completes** 100% **of its missions** at 34.54 ± 0.16 s. The previous figure was an artefact of a 35.0 s timeout inherited from an earlier, incorrectly scaled kinematic model, not a property of the algorithm. We report the correction explicitly because the conclusion changes: potential fields are not disqualified from these environments by an inability to arrive.

In our evaluation the limitation is social compliance rather than mission completion: what tells against APF here is *how* it arrives. APF records a higher intimate-space exposure than any planner evaluated except ORCA—median 10.18 person-s [IQR 8.82, 13.81], against 0.00 [0.00, 0.00] for D^{2}RO (Wilcoxon signed-rank, mean difference -11.04 person-s, 95% CI $[-11.97, -10.07]$, Holm-adjusted $p = 4.7 \times 10^{-16}$). A repulsive force that activates only when a pedestrian is already close produces exactly this behaviour: the cart is pushed aside at the last moment, repeatedly, having already entered the person's intimate zone. The deficiency is one of *anticipation*, which is precisely what a cost field evaluated over a planning horizon supplies and a local force law cannot.

2) ORCA and Decentralized Local MAPF: results we decline to over-interpret: Our implementations of Reactive ORCA and Decentralized Local MAPF complete no missions under the common arrival criterion. Instrumented traces show ORCA agents stalled for 91–99% of control ticks, having covered only a fraction of a $\sim$17 m mission, which is consistent with the velocity-obstacle infeasibility mode described in Section I-A1: in a corridor narrower than twice the safety envelope, the half-plane constraints admit no feasible velocity and the agent halts.

We deliberately stop short of the stronger claim. ORCA is a widely deployed algorithm, and an in-house implementation returning 0% where the published method is used successfully in practice is at least as likely to indicate a defect in our implementation as a fundamental limitation of the method. **We therefore report these as properties of our implementations, not of the algorithms**, and we identify validation against a reference implementation on a canonical benchmark as required work before any general claim about ORCA's behaviour in single-file corridors can be sustained. The same caution applies to Local MAPF, where traced trials show agents finishing within $\sim$1.3 m of their goals rather than livelocked—a near-miss, not the permanent token-swapping deadlock previously asserted. Consequently, none of the contributions claimed in this paper rest on the ORCA or MAPF baselines; they rest on the comparison against the matched-controller and unmatched shortest-path arms and APF, all of which complete their missions. For the same reason these two implementations are **excluded from Fig. 4**: plotting them at 0% beside the proposed method would communicate visually a comparison that this paragraph asks the reader not to draw. For completeness, both complete 0.0% [0.0, 3.7] of missions, neither records a deadlock, and their median intimate-space exposure is 11.45 person-s [0.00, 38.33] for ORCA and 6.40 [6.15, 6.60] for Local MAPF.

3) Anticipatory Horizon Extension vs. Myopic Backtracking: In decentralized navigation without peer-to-peer telemetry ($W_{\text{mesh}} = 0$), trailing carts operate under myopic range-limited sensing ($R_s = 7.2$ m). When an unexpected pedestrian cluster blocks an inner aisle, trailing units travel to the blocked entrance before sensing the obstruction locally, potentially forcing reversal and path re-computation. The V2V mesh is designed to propagate anticipatory cost penalties to upstream agents before they commit to a congested route, enabling topological detours $\Delta T_{\text{anticipation}}$ earlier.

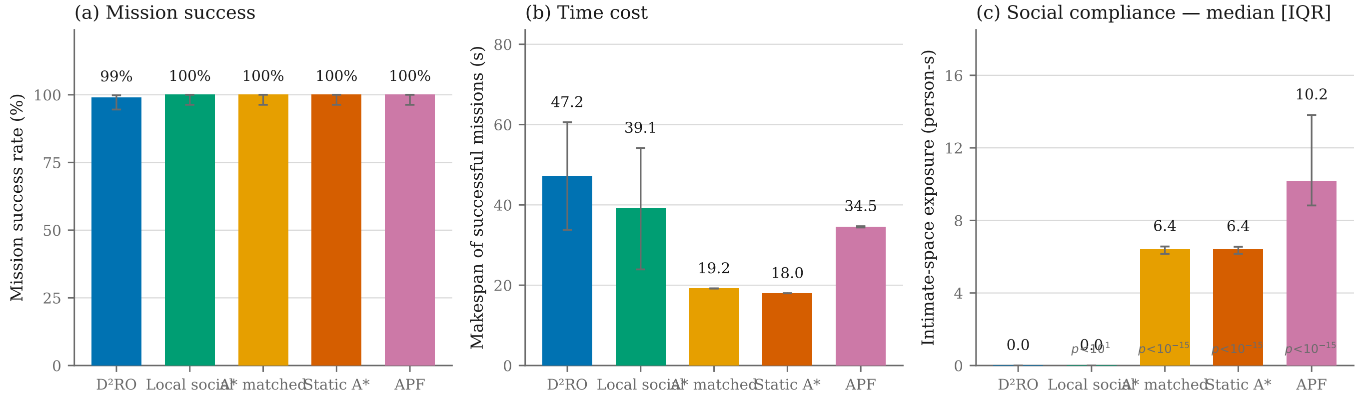

Fig. 4: Comparative evaluation across (a) mission success rate, (b) makespan of successful missions, and (c) intimate-space exposure (median, IQR whiskers). Panels (b) and (c) show the trade-off directly: D^{2}RO is slower than Static A^* and APF while intruding on personal space substantially less. Success rates in (a) are statistically indistinguishable between D^{2}RO, Static A^* and APF.

Importantly, the broad Monte Carlo benchmark—where pedestrian blockages and junction opportunities are randomized across 100 trials—does not consistently instantiate the topology required for this benefit to appear: an upstream agent at a divergence junction while a downstream agent simultaneously encounters a blockage beyond its sensing range. In those trials the mesh pays its overhead (cost-field propagation and broadcast-triggered replanning) without collecting its return, which is why the no-mesh ablation completes the benchmark faster than the full system (38.23 ± 16.88 s vs. 47.38 ± 13.96 s; Table IV). This does not invalidate the mechanism, but neither does the benchmark evidence it: a general-scenario experiment is simply not a test of anticipatory communication. Section VI-H therefore evaluates the mesh under conditions that do instantiate the required topology.

C. Comparative Benchmark: what is attributable to what

Fig. 4 does not show the proposed method dominating on every axis, and we present it as what it is. No planner recorded a corridor deadlock in any trial, so that outcome does not separate them and is not discussed further.

Social routing works, and the effect is categorical. Against planners that do not price personal space, the separation is not marginal: median intimate-space exposure is 0 person-seconds [IQR 0, 0] for D^{2}RO against 6.40 [6.2, 6.6] for both shortest-path arms and 10.18 [8.8, 13.8] for APF (Wilcoxon signed-rank, Holm-adjusted $p = 9.4 \times 10^{-16}$ and 4.7×10^{-16}). Expressed as distinct encounters, the median trial has 0 inward boundary crossings against 5 and 8 respectively.

Isolating the planner from the vehicle. A shortest-path baseline that snaps its heading at the next waypoint is not a clean comparator for a rate-limited non-holonomic planner, so we report two: a *matched-controller* arm that is a D^{2}RO agent with the social terms disabled and its route frozen, and the conventional unmatched implementation. They differ by only 1.20 s (19.20 vs. 18.00 s), so vehicle dynamics account for about 4% of the 29.18 s separating D^{2}RO from a shortest path. Both record identical exposure, confirming the matched arm is socially blind as intended.

The equal-competence control: is the distributed layer necessary here? This is the comparison Aim A2 requires and the one the literature usually omits. Local Social D* Lite carries the same proxemic cost, yielding, safety envelope and dynamic replanning as the full system, and none of its coordination machinery — no V2V mesh, no reservation. It is ordinary human-aware navigation, constructed to be D^{2}RO's equal in social behaviour so that any remaining difference is attributable to distribution alone.

It achieves *exactly* the social compliance of the full system (median 0 person-seconds; the paired difference is not distinguishable from zero, $p = 1$) at 39.06 ± 15.12 s against 47.18 ± 13.40 s, a difference that is significant ($p = 2.0 \times 10^{-5}$). In this scenario the distributed layer therefore costs roughly 8 s and returns nothing measurable, and the answer to the question is: not here.

The comparison also localises *why*. What eliminates intimate-space exposure is the socially weighted graph, which both systems share — consistent with the factorial of Section VI-D, which reaches the same conclusion by a different route. What the *distributed* layer adds appears only where its target topology is instantiated: anticipatory diversion before a blockage is observable, and corridor conflicts a single agent cannot resolve alone. Section VI-H evaluates exactly those conditions and finds the layer decisive there. Neither result subsumes the other, and the broad benchmark should not be read as evidence for or against mechanisms it does not exercise; read together they delimit when the distributed architecture is worth its cost, which is what Aim A2 set out to establish.

On makespan reporting. Fig. 4(b) summarises makespan over *successful* missions only, whereas the paired tests operate on all $N = 100$ trials including the single D^{2}RO timeout, so the two do not differ by the same amount. We report both deliberately: a successful-only descriptive answers "how long does a completed mission take?", while the all-trials paired test answers "which planner finishes sooner on the same crowd?", and a timeout is a real outcome that should not be silently dropped from the second question.

D. Separating Routing from Reactive Yielding

The comparison above still leaves an attribution question. A D^{2}RO agent both routes socially via H_{prox} and yields reactively to nearby pedestrians, so a difference against a shortest-path baseline could in principle be produced by either.

What the factor is, and what it is not. We therefore ran a 2×2 factorial, $H_{\text{prox}} \in \{\text{OFF}, \text{ON}\} \times$ yielding $\in \{\text{OFF}, \text{ON}\}$, over $N = 50$ seed-paired trials per cell. The design is deliberately narrow. The V2V mesh and the corridor reservation are disabled in *every* cell, and the dynamic D* Lite search runs in *every* cell, including those with H_{prox} off. Kinematics, collision geometry and the arrival criterion are likewise identical throughout. The consequence is that the routing factor toggles the proxemic cost term alone, rather than exchanging a frozen geometric route for the entire dynamic routing stack; an earlier version of this experiment did the latter, and could not have supported an H_{prox}-specific conclusion. Because the cells share seeds, trial t in each cell sees the same pedestrian realisation, which is what licenses the paired contrasts below.

Cell summaries cannot by themselves establish a main effect or an interaction, so Table III reports the pre-specified paired contrasts, each tested against zero across seed-matched trials and Holm-adjusted within its outcome family. Exposure is zero-inflated once H_{prox} is active — 96% of trials in Cells C and D record exactly zero — so we report medians alongside means and use a signed-rank test wherever the paired differences fail a normality check.

Total exposure must be read alongside a rate. The cells differ greatly in how long a mission lasts, from 19.20 s in Cell A to 173.89 s in Cell B, and exposure accumulates for as long as the agent is present. A difference in totals therefore mixes two distinct things: how intrusive an agent is while it is there, and how long it stays. We report both, adding exposure per minute of mission as a normalised companion to the total. Neither supersedes the other — a pedestrian crowded for fifty person-seconds is not consoled by the robot having also failed its mission — but attributing a total difference to social behaviour alone would overstate what the experiment shows. Four conclusions follow.

1. The proxemic cost term produces the social benefit. With yielding off entirely, enabling H_{prox} alone reduces exposure from a median of 6.45 person-s [IQR 6.21, 6.60] to 0.00 [0.00, 0.00] (mean difference -6.40 person-s, 95% CI $[-6.48, -6.31]$, Wilcoxon signed-rank, Holm-adjusted $p = 1.5 \times 10^{-9}$). Since the mesh and the reservation are inactive in both cells and both cells replan dynamically, this difference is attributable to H_{prox} specifically. It is not free: the same contrast costs 19.03 s of makespan (95% CI $[16.27, 22.31]$).

2. On a social route, reactive yielding gives no statistically detectable additional reduction. Adding yielding to an already social route changes exposure by 0.11 person-s (95% CI $[0.00, 0.32]$, $p = 0.59$), leaves the intrusion rate statistically unchanged ($+0.13$ person-s/min, $p = 0.59$), and leaves success unchanged (exact McNemar, zero discordant pairs, $p = 1$). This is a null result and we state it as one: the interval bounds any residual effect at well under half a person-second, but failing to detect an effect is not the same as establishing its absence.

3. Without the proxemic cost, yielding is operationally disastrous — but the harm is duration, not per-moment intrusiveness. With H_{prox} absent, enabling yielding collapses success from 100% to 12% (44 discordant pairs, all in the same direction, exact McNemar $p = 4.5 \times 10^{-13}$). An agent that halts for a pedestrian obstructing its route has no social cost gradient to steer around them, so it waits beside the obstruction until the budget expires.

Total exposure rises accordingly, by 42.92 person-s (95% CI $[37.89, 48.08]$, $p = 4.7 \times 10^{-21}$), and it would be easy to read that as the agent behaving far more intrusively. It is not. These cells differ nine-fold in how long a mission lasts (173.89 s against 19.20 s), and exposure accrues for the whole of it. Per minute of mission the intrusion rate actually *falls*, from 20.11 to 17.02 person-s/min ($\Delta = -3.09$, 95% CI $[-4.78, -1.40]$, $p = 0.002$): a robot that stops for people is, moment to moment, slightly less intrusive than one that drives through them. What yielding without a route-level social gradient destroys is mission completion, and the large total exposure is a consequence of the resulting timeouts.

4. The effect of yielding is route-context dependent, and the interaction changes sign under normalisation. On total exposure the interaction $(D - C) - (B - A)$ is strongly negative (-42.81 person-s, 95% CI $[-47.99, -37.76]$, $p = 4.7 \times 10^{-21}$); on exposure *rate* it is positive and small ($+3.22$ person-s/min, 95% CI $[1.50, 4.93]$, $p = 0.002$). Both are correct, and together they say something the total alone does not: the large negative interaction is carried by mission duration rather than by moment-to-moment behaviour.

The defensible statement is therefore that reactive yielding is context-dependent rather than beneficial-only-with-routing. Without a social cost gradient it induces prolonged obstruction and mission failure; once socially weighted routing is active it provides no detectable additional exposure reduction. In neither regime does the reactive layer substitute for the routing layer, which is the attribution this experiment was built to establish.

E. Component Ablation Analysis

The ablation in Table IV distinguishes the terms of Eq. (3) that are load-bearing in this benchmark from those that are not, and we report both outcomes.

The proxemic term carries the framework. Removing H_{prox} collapses mission success from 100.0% to 11.0% and raises the median discomfort integral from 0 [IQR 0, 0] to 13.5 [11.2, 15.6], with makespan degrading from 47.38 ± 13.96 s to 172.05 ± 26.67 s. Without a social cost the planner routes straight into crowds, then relies on the reactive yielding layer to extricate itself, which it frequently cannot do within the time budget. This single term accounts for most of what distinguishes D^{2}RO from Static A^*.

The mesh and reservation terms are not exercised by this benchmark. Removing W_{mesh} or R_{lock} leaves success at 100.0% and *reduces* makespan, from 47.38 ± 13.96 s to 38.23 ± 16.88 s and 37.86 ± 15.13 s respectively. The

TABLE II: Proxemic routing (H_{prox}) versus reactive human yielding ($N = 50$ seed-paired trials per cell). Mesh and corridor reservation are disabled in *every* cell and the dynamic D* Lite search is active in every cell, so the only quantities that vary are the two named factors. Exposure is person-seconds inside the intimate boundary, reported as median [IQR] because it is zero-inflated once H_{prox} is active, with mean ± SD alongside. Because the cells differ greatly in how long a mission lasts, total exposure confounds intrusiveness with time spent in the environment; the final column therefore normalises by mission duration, and the two should be read together.

Configuration	Success (95% CI)	Makespan (s)	Exposure median [IQR]	Exposure mean ± SD	Encounters	Exposure rate (person-s/min)
H_{prox} OFF, yield OFF	100.0% [92.9, 100.0]	19.20 ± 0.00	6.45 [6.21, 6.60]	6.43 ± 0.27	5.0 [5.0, 5.0]	20.11 ± 0.84
H_{prox} OFF, yield ON	12.0% [5.6, 23.8]	173.89 ± 17.01	48.00 [38.51, 57.86]	49.35 ± 18.38	7.0 [5.2, 9.8]	17.02 ± 6.06
H_{prox} ON, yield OFF	100.0% [92.9, 100.0]	38.23 ± 10.90	0.00 [0.00, 0.00]	0.03 ± 0.16	0.0 [0.0, 0.0]	0.04 ± 0.23
H_{prox} ON, yield ON (**D^{2}RO**)	100.0% [92.9, 100.0]	38.40 ± 11.04	0.00 [0.00, 0.00]	0.14 ± 0.80	0.0 [0.0, 0.0]	0.17 ± 0.94

TABLE III: Pre-specified paired contrasts for the 2×2 factorial ($N = 50$ seed-matched trials per contrast). A negative effect is a reduction. Each contrast is tested against zero across matched trials — by paired t-test where the differences are normal by Shapiro–Wilk and Wilcoxon signed-rank otherwise, with the test used named in the table — and p values are Holm-adjusted within the outcome family. Intervals are percentile bootstrap 95% CIs on the mean difference. Binary success is compared by exact McNemar on the same matched trials. The upper block gives total exposure in person-seconds; the lower block gives the same contrasts on exposure per minute of mission, which separates how intrusive an agent is from how long it remains in the environment. The two blocks disagree in sign for the yielding effect and the interaction, and that disagreement is the point: see Section VI-D.

Contrast	Cells	Median [IQR]	Mean (95% CI)	Test	p_{Holm}	Success p_{Holm}
Total exposure (person-s)						
H_{prox} main effect, yielding OFF	C − A	−6.45 [−6.60, −6.20]	−6.40 (−6.48, −6.31)	Wilcoxon signed-rank	1.5×10^{-9}	1
H_{prox} main effect, yielding ON	D − B	−48.00 [−57.86, −38.51]	−49.21 (−54.40, −44.16)	paired t-test	2.1×10^{-23}	4.5×10^{-13}
Yielding main effect, H_{prox} OFF	B − A	41.60 [32.31, 51.50]	42.92 (37.89, 48.08)	paired t-test	4.7×10^{-21}	4.5×10^{-13}
Yielding main effect, H_{prox} ON	D − C	0.00 [0.00, 0.00]	0.11 (0.00, 0.32)	Wilcoxon signed-rank	0.59	1
Interaction	(D − C) − (B − A)	−41.60 [−51.50, −32.31]	−42.81 (−47.99, −37.76)	paired t-test	4.7×10^{-21}	–
Exposure rate (person-s per minute of mission)						
H_{prox} main effect, yielding OFF	C − A	−20.16 [−20.62, −19.38]	−20.06 (−20.29, −19.82)	paired t-test	2.4×10^{-68}	–
H_{prox} main effect, yielding ON	D − B	−16.19 [−20.85, −13.60]	−16.85 (−18.59, −15.13)	paired t-test	8.5×10^{-24}	–
Yielding main effect, H_{prox} OFF	B − A	−3.65 [−5.98, 0.08]	−3.09 (−4.78, −1.40)	paired t-test	0.00184	–
Yielding main effect, H_{prox} ON	D − C	0.00 [0.00, 0.00]	0.13 (0.00, 0.36)	Wilcoxon signed-rank	0.59	–
Interaction	(D − C) − (B − A)	3.65 [−0.08, 5.98]	3.22 (1.50, 4.93)	paired t-test	0.00184	–

weight-sensitivity study of Section VI-I corroborates this independently: scaling w_M by any factor from 0.5 to 1.5 changes makespan by 0.0 s, because in this scenario the term it weights is almost never active. The reservation admits no such test, being a feasibility constraint rather than a weighted term; the ablation lifts it outright instead. Read alone, this ablation argues against both mechanisms. The explanation is that a randomized supermarket scenario rarely instantiates the topology each mechanism addresses—a downstream blockage outside a trailing agent's sensing range, or two agents committed to the same single-file corridor from opposite ends—while their costs (broadcast-triggered replanning, conservative corridor deferral) are paid in every trial. Section VI-H therefore evaluates both under controlled conditions that do instantiate those topologies. We consider it important to report the negative general-scenario result alongside the positive controlled one rather than only the latter.

The safety term: planner and controller separated. Earlier revisions ablated "safety" with a single switch that removed the w_S graph-cost term *and* the reactive safety controller (clearance bubble, shelf margin, following gap) at once, so the resulting effect could not be attributed to either. They are now ablated independently, and the attribution is unambiguous: removing the graph term alone raises median fixture contacts from 3 [2, 6] to 5 [5, 8], whereas removing the reactive controller alone raises them to 48 [35, 69], and removing both to 93.5 [82, 96].

The effect is therefore mostly the *controller*, not the cost term. Pricing clearance in the graph helps, but only modestly; what actually keeps the chassis off the fixtures is the reactive clearance enforcement in the motion layer. We state this plainly because the previous full-stack ablation invited the opposite reading, and because it bounds what the w_S term should be credited with.

Counts are reported as median [IQR] rather than $\mu \pm \sigma$: they are right-skewed enough that the standard deviation exceeds the mean in several configurations, and a summary such as 5.71 ± 7.29 implies negative contacts, which cannot occur.

F. Cross-Domain Failure Mode Analysis

Aim A3 requires that the planner transfer across dissimilar architectures without retuning. Table V supports this: with identical weights and no domain-specific parameters, D^{2}RO completes 99.0% of missions in the retail supermarket, 100.0% in the clinical hospital and 95.0% in the airport terminal. Cross-topology transfer within the evaluated simulation family is therefore demonstrated: the three domains are hand-designed layouts sharing one physics abstraction and one sensing model, so this establishes that the planner does not depend on a tuned topology, not that it would transfer to a real facility. What differs across domains is not reliability but the *character* of the routing effort, which the remaining columns expose.

The hospital is the easiest domain, not the hardest. It records the highest success rate and the lowest replan count

TABLE IV: Component ablation in the retail supermarket domain ($N = 100$ trials per configuration). The upper block removes one soft cost term at a time from Eq. (1), or lifts the reservation feasibility constraint; the lower block separates the kinematic safety *cost term* from the reactive safety *controller*, which are distinct mechanisms. All other parameters are held fixed. Counts are reported as median [IQR] because they are right-skewed; makespan is mean ± SD. No configuration recorded a deadlock in any trial.

Configuration	Success (95% CI)	Makespan (s)	Discomfort	Fixture contacts	Contact ticks
Routing-cost ablations					
Full D^{2}RO	100.0% [96.3, 100.0]	47.38 ± 13.96	0.0 [0.0, 0.0]	3 [2, 6]	154 [114, 191]
w/o V2V mesh W_{mesh}	100.0% [96.3, 100.0]	38.23 ± 16.88	0.0 [0.0, 0.0]	3 [3, 9]	131 [80, 246]
w/o proxemics H_{prox}	11.0% [6.3, 18.6]	172.05 ± 26.67	13.5 [11.2, 15.6]	56 [39, 78]	219 [127, 313]
reservation constraint R_{lock} lifted	100.0% [96.3, 100.0]	37.86 ± 15.13	0.0 [0.0, 0.0]	3 [3, 9]	131 [80, 242]
Safety attribution					
Full D^{2}RO	100.0% [96.3, 100.0]	47.38 ± 13.96	0.0 [0.0, 0.0]	3 [2, 6]	154 [114, 191]
$w_S = 0$, reactive controller retained	99.0% [94.6, 99.8]	53.52 ± 24.71	0.0 [0.0, 0.0]	5 [5, 8]	270 [245, 313]
controller off, w_S retained	100.0% [96.3, 100.0]	42.34 ± 9.84	0.0 [0.0, 0.0]	48 [35, 69]	150 [125, 173]
both off	100.0% [96.3, 100.0]	43.02 ± 8.84	0.0 [0.0, 0.0]	94 [82, 96]	244 [224, 310]

TABLE V: Cross-domain generalisation of the unchanged planner across three topologically distinct environments ($N = 100$ trials per domain). Intimate exposure is reported as median [IQR]; the distribution is zero-inflated.

Domain	Success (95% CI)	Makespan (s)	Mean transit (s)	Intimate exposure	D* Lite replans
Retail Supermarket	99.0% [94.6, 99.8]	49.89 ± 19.94	25.90 ± 5.57	0 [0, 0]	502.3 ± 151.9
Clinical Hospital	100.0% [96.3, 100.0]	46.12 ± 10.12	31.78 ± 3.89	0 [0, 0]	351.9 ± 65.5
Airport Terminal	95.0% [88.8, 97.8]	74.63 ± 35.71	36.84 ± 15.01	62 [18, 130]	989.4 ± 361.0

(351.9 ± 66.0), despite having the narrowest corridors. Its topology is strongly channelised: routes are largely predetermined by the corridor structure, so there are few alternatives to evaluate and little for the planner to reconsider once committed. Its longer mean transit (31.78 ± 3.89 s) reflects distance and mandatory alcove yields, not indecision.

The airport is the hardest, and for an instructive reason. It requires 989.4 ± 369.7 replans—nearly twice the supermarket's 502.3 ± 155.6—and is the only domain with non-zero median intimate exposure (62 [18, 130], against 0 [0, 0] in both other domains). An open concourse offers a large, continuously changing set of admissible routes, so a crowd shifting a few metres alters the cost field enough to justify a new plan; the planner spends its effort re-deciding rather than executing. The wide interquartile range indicates the exposure is concentrated in trials where the fleet must cross a dense gate-area crowd, rather than being incurred uniformly. Open topologies thus stress the framework differently from constrained ones—the difficulty is choice abundance, not scarcity—which motivates crowd-density-adaptive scheduling of w_H and w_M, and a replan-hysteresis term, as future work.

We note that these figures supersede an earlier revision reporting 80.0% airport success. In that revision the airport scenario reset its random seed to a constant, so its 100 nominal trials were 100 repetitions of a single trial; the defect is corrected and the domain now varies per trial as the other two do.

G. Decoupled Scalability Analysis

We vary crowd density and fleet size independently ($N = 100$ trials per point) so that the two effects are not confounded. The two dimensions behave very differently, and only one of them is benign.

Crowd density scales benignly (Fig. 5). Increasing pedestrians from 2 to 30 raises the control-step compute time only from 0.093 ms to 0.172 ms and makespan from 30.10 s to 56.30 s, while success falls just 100.0% → 98.0%. Incremental repair is doing its job: a denser crowd perturbs more edges, but D* Lite repairs only the affected subgraph, so cost grows with the perturbation rather than with the map. Even at the highest density the controller consumes under 0.4% of the 50 ms budget.

A caveat on the latency figures specifically. Replan latency is the only quantity reported in this paper that is not deterministic. Every other metric—success, makespan, exposure, packet counts—is a property of the seeded simulation and reproduces bit-identically on reruns; latency is wall-clock time and therefore depends on the machine, its load, and the Python interpreter. Regenerating the datasets on a more heavily loaded machine moved these figures by roughly a factor of two while leaving every other number in this paper unchanged. The correct reading is consequently ordinal, not absolute: incremental repair costs a small and slowly growing fraction of the control budget, with ample headroom on commodity hardware. A precise per-platform figure would require a dedicated timing harness on quiesced hardware, which we do not claim here.

Fleet size does not (Fig. 6), and this is the framework's clearest limitation. Success holds near 100% up to 8 carts, then falls to 85.0% at 10 and 78.0% at 12, while makespan more than doubles (50.99 s → 110.17 s). The mechanism is visible in the mesh column: broadcast volume grows super-linearly, from 8.50 ± 5.72 packets at two carts to 1158.60 ± 336.83

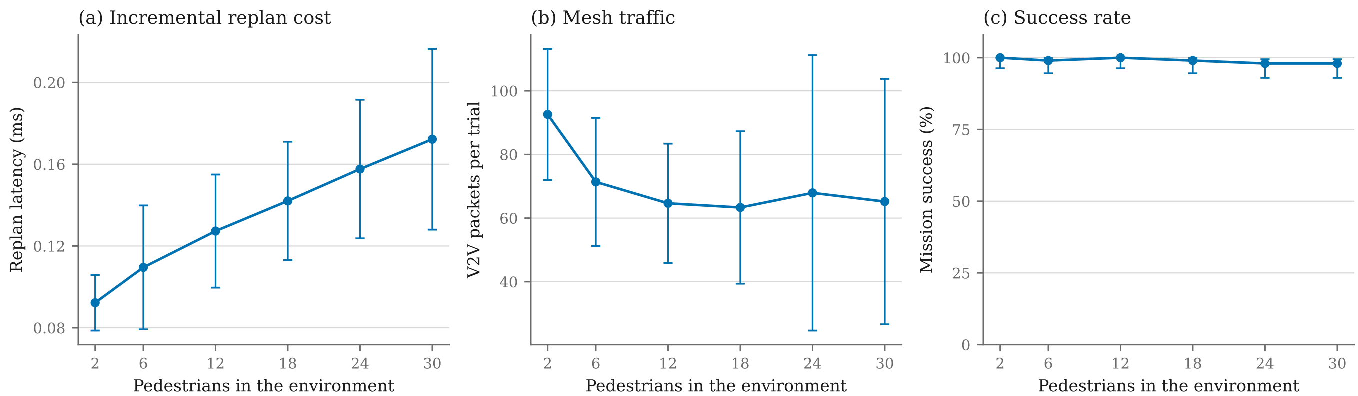

Fig. 5: Crowd-density scalability at a fixed fleet of four carts. Error bars are ± 1 SD, except (c) which shows the Wilson 95% interval.

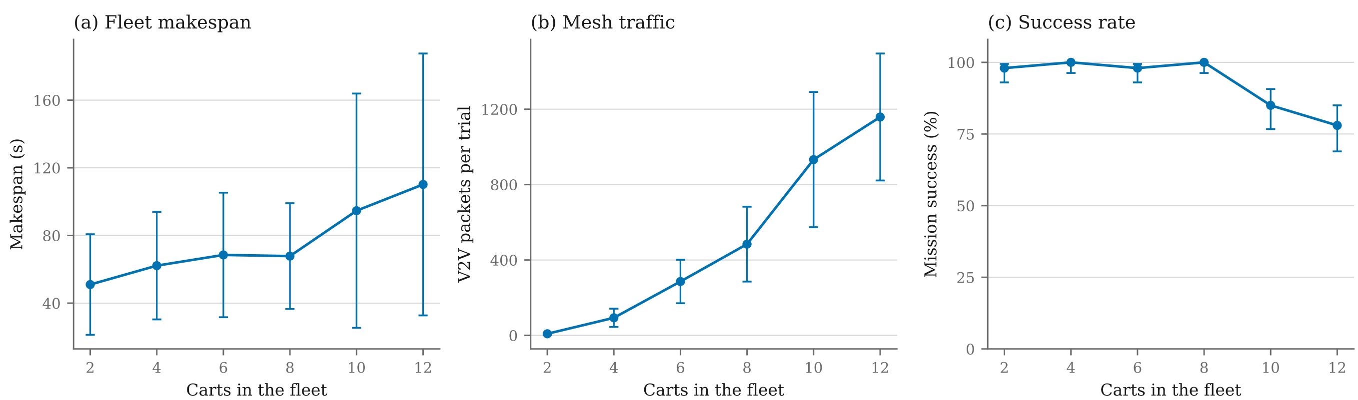

Fig. 6: Fleet-size scalability at a fixed crowd of ten pedestrians. Error bars are ± 1 SD, except (c) which shows the Wilson 95% interval.

at twelve, because every agent broadcasts observations that every other agent must incorporate. Beyond roughly eight carts in a 36×24 m floorplan the fleet begins to congest itself: agents increasingly replan around each other's reservations and telemetry rather than around pedestrians.

We state this bound plainly rather than describing the degradation as graceful. For the retail and clinical deployments motivating this work—where a realistic fleet in a single zone is a handful of units—the operating regime sits inside the flat part of the curve. But the present design should not be represented as scaling to large fleets in a confined space, and mesh-traffic suppression (relevance filtering, spatial scoping of broadcasts) is the natural next step for that regime.

H. Mechanism-Specific Validation

Because the broad Monte Carlo benchmark does not reliably trigger the specific topological conditions under which V2V mesh anticipation and the corridor reservation protocol provide their primary contribution, we designed two controlled experiments that directly exercise each mechanism. These complement the general-scenario evaluation and provide targeted evidence for the core algorithmic novelties.

1) Experiment A: V2V Mesh Anticipation ($N = 50$ trials): Two carts traverse the same aisle system, the leader ahead of the follower, and a pedestrian cluster blocks the leader's forward aisle mid-route. The precondition the implementation actually enforces is stated precisely: a trial is admitted only if **the blockage lies outside the follower's** 7.2 **m sensing radius at the moment it is created**, so the follower cannot have perceived it directly. (An earlier description of this experiment claimed the follower begins "10+ m behind the leader"; that is not the condition the code tests, and the separation between carts is not what makes the trial admissible.) An alternative divergence junction exists upstream of the follower, where it can reroute *before* committing to the congested path. We compare **Mesh ON** against **Mesh OFF** on identical seeds.

Two definitions matter for reading Table VI. *Anticipation lead time* is the interval between the follower's diversion and the moment the blockage would have entered its sensing radius; it is negative when the diversion happens only after contact. *Backtracking distance* is the cumulative increase in Euclidean distance to the goal, i.e. ground covered in the wrong direction, rather than a count of re-traversed edges. We also note that sensing here is **range-limited rather than line-of-sight**: pedestrians are selected by Euclidean distance without ray-casting against shelf occlusion, so the follower's perceptual advantage is if anything overstated relative to a real sensor behind a gondola.

TABLE VI: Mechanism experiment A: V2V mesh anticipation. A corridor is blocked outside the follower's onboard sensing radius; with the mesh enabled the follower learns of the obstruction before observing it ($N = 50$ paired trials; the same seed drives both arms, so each row is a within-pair comparison). Continuous outcomes use the Wilcoxon signed-rank test; rows whose two arms were equal in every pair admit no test. p-values are Holm-adjusted across the rows of this table. Both arms completed 100.0% of missions, so mission success does not separate them and is omitted from the table.

Metric	ON	OFF	p (Holm)
Anticipation lead time (s)	10.70 ± 4.20	-0.10 ± 0.04	3.8×10^{-9}
Backtrack distance (m)	1.08 ± 0.68	2.73 ± 0.87	3.2×10^{-8}
Path length (m)	23.97 ± 3.58	25.81 ± 3.44	0.00283
Makespan (s)	21.44 ± 3.40	23.78 ± 3.61	0.00196

TABLE VII: Mechanism experiment B: directional corridor reservation. Two carts are routed into the same single-file corridor from opposite ends ($N = 50$ paired trials; the same seed drives both arms, so each row is a within-pair comparison). Continuous outcomes use the Wilcoxon signed-rank test; rows whose two arms were equal in every pair admit no test. p-values are Holm-adjusted across the rows of this table.

Metric	ON	OFF	p (Holm)
Head-on encounters	1.08 ± 1.18	1.00 ± 0.00	1
Deadlocks	0.00 ± 0.00	0.00 ± 0.00	1
Lock wait, total (s)	0.03 ± 0.07	0.00 ± 0.00	0.209
Off-corridor vertices	2.16 ± 1.36	0.00 ± 0.00	2.3×10^{-8}
Route reconsiderations	25.2 ± 3.2	10.2 ± 4.1	6.5×10^{-9}
Corridor occupancy (s)	40.01 ± 30.87	89.41 ± 42.42	0.00484
Makespan (s)	40.05 ± 30.85	89.42 ± 42.39	0.00484
Mission success (McNemar)	88.0%	36.0%	1.7×10^{-4}

The mechanism behaves as designed. With the mesh enabled the trailing cart diverts 10.70 ± 4.20 s before it could have observed the obstruction itself, against -0.10 ± 0.04 s with the mesh disabled—that is, without telemetry the diversion occurs only *after* contact with the blockage. Backtracking falls from 2.73 ± 0.87 m to 1.08 ± 0.68 m and makespan from 23.78 ± 3.61 s to 21.44 ± 3.40 s (Wilcoxon signed-rank, Holm-adjusted $p = 3.8 \times 10^{-9}$ and 2.0×10^{-3} respectively). Success is 100% in both arms: the mesh does not determine *whether* the mission completes here, it determines how much wasted travel is incurred, which is the claim the mechanism was designed to support. Together with the ablation result of Section VI-C, the honest summary is that anticipatory telemetry pays when the topology affords an early alternative and costs a little when it does not.

2) Experiment B: Directional Corridor Reservation ($N = 50$ trials): Two carts approach the same single-file corridor from opposite ends, with randomised arrival offsets. The corridor is verified to be genuinely single-file before each trial runs. We compare **Lock ON** against **Lock OFF** on identical seeds.

The effect is large. Mission success rises from 36.0% to 88.0% (McNemar exact, Holm-adjusted $p = 1.7 \times 10^{-4}$) and corridor occupancy falls from 89.41 ± 42.42 s to 40.01 ± 30.87 s ($p = 4.8 \times 10^{-3}$). Without the protocol, the majority of trials fail.

The mechanism is not mutual exclusion. A reader who sees "mutex" will expect one agent to hold the corridor while the other queues, and that is not what happens. Head-on encounters are statistically indistinguishable between arms (1.08 ± 1.18 with the protocol, 1.00 ± 0.00 without; $p = 1$), so the protocol does not prevent the encounter. The deadlock counter registers zero in *both* arms, so the $N_{\text{deadlock}} = 0$ result cannot be credited to it either. And total accumulated waiting is 0.030 ± 0.070 s against 0.000 s ($p = 0.21$): agents essentially never queue.

What happens instead is cost-projected diversion. When an agent observes a peer's stronger claim, that claim enters its *graph* as an effectively infinite edge cost, and D* Lite reroutes through a parallel aisle. The released dataset records this directly. With the protocol enabled, agents occupy 2.16 ± 1.36 vertices outside the contested corridor and issue 25.18 ± 3.22 route reconsiderations; with it disabled they occupy 0.00 such vertices and issue 10.16 ± 4.12 (Holm-adjusted $p = 2.3 \times 10^{-8}$ and 6.5×10^{-9}). The corridor is vacated by re-routing, not by turn-taking, which is consistent with occupancy halving while wait time stays at zero.

The head-on encounter still occurs because it is what triggers the exchange of claims; what the protocol changes is the outcome, not the frequency. Correspondingly, the failure mode without it is not a deadlock in the strict sense but a persistent mutual obstruction that expires against the time budget, which is why a counter keyed on unroutable planner states never fires in either arm.

We therefore avoid the term "mutex" for this component and call it a *directional reservation with cost-projected diversion*. That is a stronger and more general claim than mutual exclusion — peer intent is converted into a route-level cost that any graph planner can act on — but it is a different claim, and the stricter computer-science reading of "mutex" is not what the evidence supports.

A note on the wait metric. An earlier revision reported 0.00 ± 0.00 s of waiting and used it as evidence for this interpretation. That figure was not measurable: the per-episode wait timer was reset when the corridor was released, before the experiment read it at end of run, so it would have read zero however long agents actually waited. The value quoted above comes from a separate accumulator that release does not reset. The conclusion survives the correction, but it now rests on a metric that could have contradicted it.

I. Weight Sensitivity

The proposed cost function is a weighted sum over four soft terms, so its behaviour could in principle depend delicately on the four cost weights. If the reported operating point sat on a narrow ridge, the result would be fragile in a way a reader must be told about. We therefore perturb each weight in turn by $\times\{0.5, 0.75, 1.25, 1.5\}$ while holding the other three at nominal, and evaluate on a seed set **disjoint** from every other experiment in this paper: the weights were selected by hand, and assessing their robustness on the seeds used to select and report them would be a form of tuning on the evaluation set.

The operating point is a plateau, not a ridge. Mission success remains between 97% and 100% under every perturbation of every weight. No setting in the grid produces a collapse,

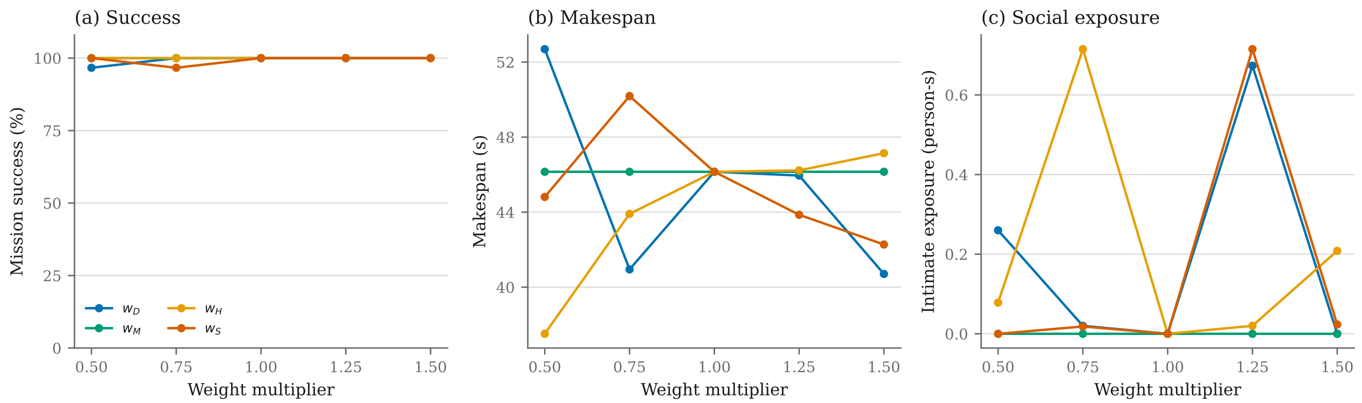

Fig. 7: Sensitivity of the four soft cost weights (w_D, w_M, w_H, w_S). Each weight is scaled in turn while the other three are held at nominal; the $\times 1.0$ point is the shared nominal configuration. Corridor reservation is a hard feasibility constraint and is evaluated separately. Evaluated on a seed set disjoint from every other experiment.

and the nominal configuration is not a special point on which the method depends.

Makespan responds to the social weight in the expected direction, and to the distance weight less regularly than we previously reported. Increasing w_H, which prices proximity to people, lengthens missions monotonically (37.5 s at $\times 0.5$ to 47.1 s at $\times 1.5$): that is the distance-versus-sociability dial the formulation is meant to expose, and a deployment wanting faster, less deferent carts can move along it. The w_D response is not monotone (52.7, 40.9, 46.2, 46.0, 40.7 s across the grid), which we attribute to route switching — at some weightings the planner commits to a structurally different corridor rather than the same corridor traversed at a different price.

An earlier revision reported a clean monotone w_D trend. That result is superseded and should not be relied upon: it was computed with a heuristic that is inadmissible for $w_D < 1$ (Section 1), so the planner was returning suboptimal paths in precisely the configurations that anchored the trend.

w_M **does nothing here, and that is informative.** Scaling the mesh weight by any factor in the grid changes makespan by 0.0 s, corroborating the ablation and the benchmark: in a randomised supermarket scenario the mesh term is almost never active, so its coefficient has nothing to scale. The reservation is absent from this study altogether, because it is a feasibility constraint rather than a weighted term and has no magnitude to perturb (Section III-B4); it is evaluated by the ON/OFF mechanism experiment instead.

Consistent with this, we describe $[w_D, w_M, w_H, w_S] = [1.0, 1.5, 2.0, 1.2]$ throughout as **hand-selected nominal weights**. No formal calibration procedure was performed, and we make no claim that they are optimal — only that the method's behaviour is insensitive to their exact values over the range tested.

J. Communication Robustness

Every experiment reported above assumes an **ideal channel**: the V2V mesh delivers every packet with zero delay. Since the deployment discussion appeals to RF attenuation in steel-fixture retail environments, that assumption should be tested rather than left implicit.

We sweep packet loss $\in \{0, 5, 10, 20\}\%$ against one-hop latency $\in \{0, 50, 100, 200\}$ ms in the broad scenario, a full factorial of sixteen conditions at $N = 30$ trials each. There, mission success is 100% in every condition and makespan varies only between 43.6 and 47.2 s — a spread far smaller than the $\sim$16 s standard deviation within any single condition.

That null result is weak evidence, however, and we do not lean on it: the broad scenario barely uses the mesh, so degrading a channel that carries little traffic naturally changes little. **The informative test is to degrade the channel where the mesh is causally responsible for the outcome**, so we repeat Mechanism A (Section VI-H) across a packet loss $\in \{0, 10, 20\}\%$ by one-hop latency $\in \{0, 100, 200\}$ ms grid, $N = 20$ seed-matched Mesh-ON/Mesh-OFF pairs per condition.

The quantity we analyse is the paired effect, not the absolute arm value. Every channel condition runs a Mesh-ON and a Mesh-OFF trial on the same seed, so the estimate the design supports is the within-pair difference $\Delta T_{\text{anticipation}} = T_{\text{Mesh ON}} - T_{\text{Mesh OFF}}$; quoting the Mesh-ON lead time alone would discard the control the experiment was built to provide. Fig. 8 plots that difference, with bootstrap intervals, against loss for each latency level. The mesh retains a significant advantage in *every* condition tested — no interval approaches zero — so degradation here erodes the advantage rather than eliminating it.

Whether the advantage *changes* with the channel is a separate question, and one that per-condition effects cannot answer, so we contrast each channel against the clean one, pairing on trial and on the other factor. Relative to a clean channel, the mesh advantage falls by 0.80 s at 10% loss — a change the signed-rank test does not detect (Holm-adjusted $p = 0.16$) and a small one against an advantage of roughly 11 s — and by 4.26 s at 20% loss (95% CI $[-5.73, -2.90]$, $p = 6.6 \times 10^{-6}$), an erosion of about 38%. One-hop delay produces a statistically detectable but practically negligible reduction over the range tested: 0.45 s at 100 ms and 0.55 s at 200 ms (both $p \leq 1.0 \times 10^{-4}$).

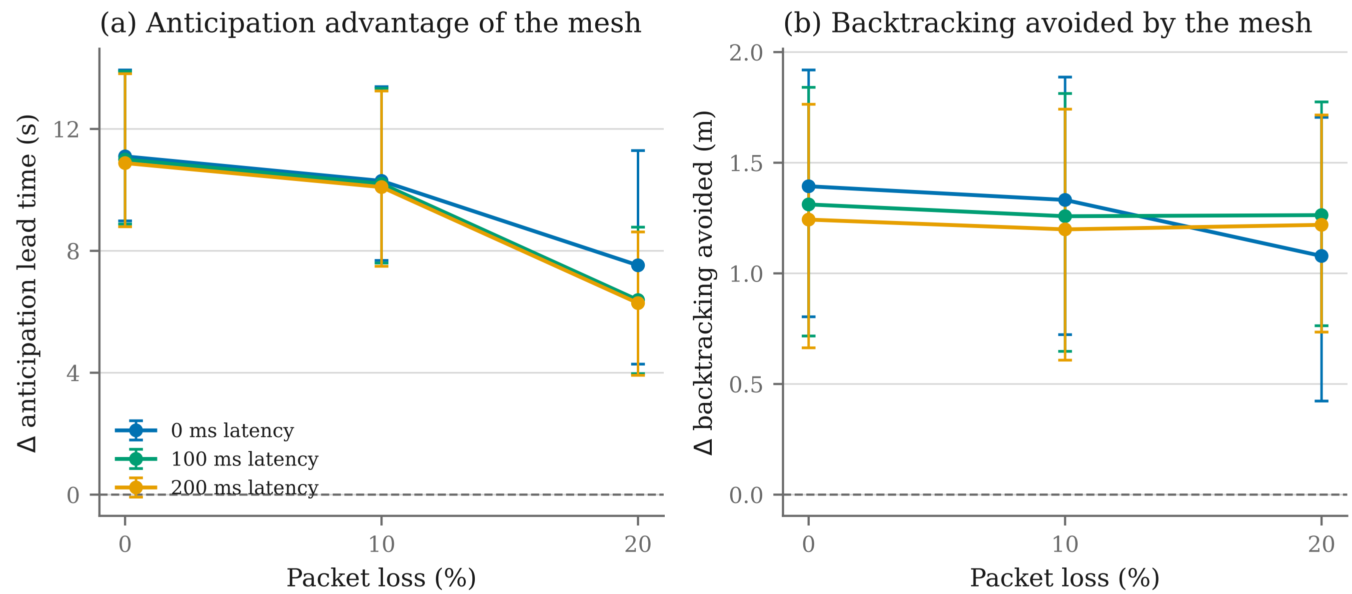

Fig. 8: Paired effect of the V2V mesh on Mechanism A as the channel degrades ($N = 20$ seed-matched Mesh-ON/Mesh-OFF pairs per channel condition). Each point is the within-pair difference, not an absolute arm value, so the controlled design of the mechanism experiment is preserved; error bars are percentile bootstrap 95% intervals on the mean difference. The dashed rule marks no mesh advantage.

We therefore retain the 10% figure, but as a claim with a stated basis rather than an eyeballed threshold: at 10% loss the degradation is neither statistically detectable nor practically material, while at 20% it is both. Latency is well tolerated and loss is what erodes the advantage, which is consistent with the design, since an alert that is delayed still arrives while an alert that is dropped does not.

Two caveats bound that claim. First, the mesh contributes little in this scenario to begin with (Section VI-I), so a result showing that degrading it changes little is correspondingly weak evidence: the honest reading is that degradation does no *harm*, not that the protocol would survive degradation in a scenario where the mesh carried the outcome. Second, the model is a per-hop delay and an independent drop probability; it does not reproduce bursty loss, correlated shadowing behind a metal gondola, or contention collapse under load. A physical deployment would need measurements from the actual RF environment.

We note that this experiment is only meaningful because of a correction made during this revision. The mesh implementation modelled latency via per-packet delivery timestamps, but the agent fetched its inbox without supplying the current simulation time, so every packet was released immediately regardless of its timestamp. Latency was configurable and inert. The results above were obtained after that path was repaired.

VII. Conclusion & Future Research Directions

This paper introduced the Distributed Dynamic Route Optimization (D^2RO) framework powered by Socially-Weighted Distributed Graph Optimization (SW-DGO), formalizing a 4-component soft edge traversal cost function $C_i(e,t) = w_D \cdot C_{\text{geom}}(e) + w_M \cdot C_{\text{mesh}}(e,t) + w_H \cdot C_{\text{social}}(e,t) + w_S \cdot C_{\text{kinematic}}(i,e,t)$ minimised subject to a directional corridor reservation constraint (R_{lock}) and solved by incremental D^* Lite heuristic graph repair. We assess the outcome against the four aims set out in Section I-D:

A1 (social compliance) is met, and now attributably so. Median intimate-space exposure is 0.00 person-seconds [IQR 0, 0] against 6.40 for a matched-controller shortest path and 10.18 for artificial potential fields (Holm-adjusted $p = 9.4 \times 10^{-16}$ and 4.7×10^{-16} respectively). The 2×2 factorial of Section VI-D establishes that this is produced by the socially weighted graph (H_{prox}) and not by the reactive yielding layer: routing alone drives exposure to 0.00 person-seconds, and the intrusion rate from 20.11 to 0.04 person-s/min. Adding yielding to an already social route changes exposure by 0.11 person-s ($p = 0.59$), a null we report as such. Without the proxemic cost, yielding raises total exposure by 42.92 person-s (95% CI $[37.89, 48.08]$) and collapses success to 12%, but that total reflects missions lasting nine times longer rather than more intrusive behaviour: per minute the rate falls slightly (-3.09 person-s/min, $p = 0.002$). The harm is operational, and we are careful not to claim otherwise. The compliance is bought with 19.03 s of makespan (95% CI $[16.27, 22.31]$), measured with yielding held constant.

A2 (distributed coordination, and the conditions under which it is warranted) is met, and the answer is conditional. This aim asked not whether a distributed layer *can* coordinate a fleet but when it is *needed*, which is answerable only against a control of equal social competence. Against that control the layer contributes nothing detectable in the broad scenario: an ordinary human-aware planner carrying the same proxemic cost but no mesh and no reservation matches the full system's social compliance exactly ($p = 1$) while running 8 s faster. Under the topologies the mechanisms target it is decisive, advancing rerouting by 10.70 ± 4.20 s and raising corridor success from 36.0% to 88.0% by cost-projected diversion rather than by queueing.

The distributed layer is therefore not a general improvement but a targeted one, and we regard locating that boundary as the more useful result. A deployment whose topology routinely produces blockages outside a follower's sensing range, or single-file corridors entered from both ends, should pay for the radio; one that does not would be right to omit it and run the local planner, which is simpler, faster and equally well behaved around people. Reporting the condition is what makes the positive result actionable rather than merely favourable.

A3 (generality and real-time feasibility) is met, within a stated fleet bound. The unchanged planner achieves 99.0%, 100.0% and 95.0% success across supermarket, hospital and airport topologies. Incremental repair costs a median of 0.002 ms (p95 0.10 ms) inside a 0.118 ms control step *on the evaluation host*, under 0.25% of the 50 ms budget. These are wall-clock figures and are not portable across machines; the claim they support is ordinal — repair costs microseconds, not milliseconds. The paired anticipation advantage is statistically unchanged at 10% packet loss and at one-hop latency up to 200 ms, and falls by about 38% at 20% loss. Success falls to 78.0% at twelve carts in a 36×24 m floorplan as mesh traffic grows super-linearly, and we state that ceiling as a limitation rather than describing it as graceful degradation.

A4 (attributable evaluation) is met. The matched-controller arm shows that 1.20 s of the 29.18 s gap is vehicle dynamics; the factorial separates routing from yielding; the split safety ablation separates the w_S cost term from the reactive clearance controller, and shows the latter dominates; and the sensitivity study shows the operating point is a plateau. Each reported effect is therefore a property of an identified component rather than of the system as a whole.

A. Theoretical Guarantees & Their Empirical Limits

1) **Heuristic Admissibility & Incremental Optimality:** The heuristic must be a lower bound on true cost. Since every term of Eq. (3) after the first is non-negative, a valid bound is obtained by retaining only the distance term at its smallest coefficient, $h(u, v) = \min(w_D) \|\mathbf{p}_u - \mathbf{p}_v\|$. The scaling is not cosmetic: plain Euclidean distance is admissible only while $w_D \geq 1$, and at $w_D = 0.5$ an unobstructed edge costs $0.5d$ while the unscaled heuristic claims d, so $h > c^*$ and the search may return a suboptimal path. Since the sensitivity study of Section VI-I evaluates w_D at 0.5 and 0.75, the unscaled heuristic was inadmissible for exactly those configurations, and results previously reported for them are superseded. With the scaled bound, D* Lite extracts an optimal path with respect to the currently observed cost field while recomputing only the perturbed subgraph [8]. We verify this against fresh Dijkstra solutions over 150 randomised scenarios comprising 750 successive repairs, *and* across the whole w_D grid, with a direct assertion that $h(u, g) \leq c^*(u, g)$ at every vertex.

2) **Conflict Resolution in Single-File Bottlenecks:** Directional reservations broadcast as `LOCK_REQUEST` packets inflate the contested edge cost for the deferring agent, which then diverts through a parallel aisle or holds in a Turnout Alcove (V_{alcove}). We claim empirical conflict *resolution*, not deadlock freedom: no formal guarantee is established here, and the zero deadlock counts we measure occur in both the lock-enabled and lock-disabled arms, so they cannot be attributed to the protocol. A liveness proof under bounded latency and message loss remains open.

B. Limitations

We state the principal limitations directly.

(i) The pedestrian model is simple and non-reciprocal. Simulated humans alternate between walking and browsing, choose stochastic targets and avoid fixtures, but they do not perceive or react to the robots as social agents: they neither yield, nor slow, nor adjust their path in anticipation. This study therefore demonstrates socially weighted robot navigation *around* synthetic moving pedestrians, not reciprocal human–robot interaction, and the reported exposure figures should be read as an upper bound on what a real crowd — which would partly step aside — would produce. Stronger claims would require recorded pedestrian trajectories, a reciprocal social-force model, or human-in-the-loop trials.

(ii) The weights are hand-selected. Section VI-I shows the result is insensitive to them over $\times[0.5, 1.5]$, but no calibration procedure was performed and we do not claim optimality. The study is also one-at-a-time; joint or multivariate interactions are untested.

(iii) We report when the distributed layer helps, but not how often that happens. That it is neutral-to-costly in the broad benchmark and decisive in the targeted mechanism experiments is a finding rather than a caveat (Aim A2). The genuine gap is a base rate: we did not instrument how frequently the enabling topologies — a blockage beyond a follower's sensing radius on its committed route, or a single-file corridor entered from both ends — actually arise, either in our randomised scenario or in a real facility. Without that rate a deployer cannot convert our per-event effects into an expected benefit, and the honest summary remains conditional rather than quantitative. Measuring it is the most useful single extension of this work.

(iv) ORCA and Local MAPF are our own implementations and return 0% success; no claim in this paper depends on them, and they are excluded from the primary comparison.

(v) All results are simulation-based. No physical fleet was deployed, and localisation is assumed rather than estimated.

(vi) Fleet scaling is bounded as described above, and **(vii)** sensing is range-limited rather than occlusion-aware: pedestrians are selected by Euclidean distance without ray-casting against shelves, so the perceptual advantage of a trailing agent is if anything overstated.

C. Considerations for a Physical Deployment

The evaluation in this paper is entirely simulation-based. The following are therefore *requirements a physical deployment* *would have to satisfy*, not properties of the implemented system; none of the hardware described below forms part of the evaluated framework.

1) **RF Signal Attenuation & Steel Fixture Multipath:** Metallic retail shelf gondolas and merchandise packaging attenuate 2.4 GHz RF signals appreciably. We attach no figure in dB to that attenuation because we have measured none, and the mesh model evaluated here is not calibrated against any RF environment. A deployment would likely require Sub-GHz ($868/915$ MHz) or dedicated short-range (5.9 GHz IEEE 802.11p / DSRC) links with multi-hop TTL forwarding. The exponential penalty decay $W_{\text{mesh}}(t) = W_0 \exp(-\lambda t)$ that *is* implemented and evaluated here provides the relevant safeguard in principle: stale telemetry expires on its own, so intermittent packet loss cannot permanently poison the routing graph. Our mesh model includes latency and loss, but has not been validated against measurements from a real RF environment.
2) **Indoor Positioning & Sensor Fusion:** All results assume the pose is known. A physical cart would need to recover it—plausibly by fusing wheel encoders, an IMU, Ultra-Wideband trilateration and LiDAR scan-matching through an Extended Kalman Filter [18], [20]—and the resulting localisation error would propagate into both the proxemic field and the corridor reservations. Quantifying that sensitivity is future work.
3) **Payload Invariance & Non-Holonomic Kinodynamics:** Service carts undergo large payload shifts (on the order of 15 kg empty to 65 kg loaded), altering the centre of gravity and rotational inertia. The evaluated system uses fixed kinodynamic limits; a deployment would need to modulate $a_{\max}$ and $\kappa_{\max}$ from load sensing to avoid wheel scrubbing or tip-over. This is not implemented or evaluated here.

D. Future Research Directions

The limitations above define the immediate agenda; the last two items are longer-range.

1) **Joint Weight Calibration:** Section VI-I perturbs each weight one at a time and finds a plateau, but says nothing about interactions between them. Future work should treat calibration as a multi-objective problem — a Pareto front over makespan and social exposure, task-dependent weightings, or automatic selection from a target compliance level — rather than a one-at-a-time sweep.
2) **Baseline Validation:** Reproducing our ORCA implementation's behaviour against a reference (e.g. RVO2) on a canonical benchmark, so that comparative claims rest on validated baselines.
3) **Mesh Traffic Suppression:** Relevance filtering and spatial scoping of broadcasts, to lift the fleet-size ceiling identified in Section VI-G.
4) **Heading-Augmented Search:** Implementing α_{turn} over a state space augmented with heading, allowing turn cost to enter the graph search rather than only the motion layer.
5) **Hybrid Learning-Guided Search:** Coupling Graph Neural Networks or deep reinforcement learning (e.g., PRIMAL / Learn to Follow) for global sub-goal allocation with the deterministic D^{2}RO engine for kinodynamic execution.
6) **Heterogeneous Multi-Agent Ecosystems:** Extending SW-DGO to mixed fleets—delivery pods, floor-scrubbing AGVs, human-operated wheelchairs—via vehicle-specific agility weights in the R_{lock} reservation tuple.

Data and Code Availability

All simulation source code, experimental harnesses, test suites, raw CSV datasets and vector figures will be released under open licences in a public repository; the URL is withheld here because it identifies the authors, and will be supplied on acceptance. Every number, table and figure in this paper is generated programmatically from the committed raw data by a single analysis pipeline; none is entered by hand. The results reported here correspond to release `v4.0-review4`, commit `23ab6bd`. The stamp names the commit whose sources produced this document; the compiled PDF is committed immediately afterwards, so the file in the repository is one commit later than the stamp it carries. Continuous integration rebuilds the same artefact from the tag on a clean checkout, and `paper/scripts/release_gate.py` certifies either build, refusing any in which a dataset is stale, a prose claim disagrees with the data, a reference is unresolved, or a quoted p-value does not appear in the analysis output. Each dataset carries a provenance stamp recording the SHA-256 fingerprint of the simulation source that produced it, and the analysis pipeline refuses to emit a table or figure for any dataset whose fingerprint does not match the committed code.

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